



## 5105 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

Cycle: 3, Proposal Category: GO

### INVESTIGATORS

| <i>Name</i>                                | <i>Institution</i>                                  |
|--------------------------------------------|-----------------------------------------------------|
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| Dr. Feige Wang (CoI)                       | University of Michigan                              |
| Prof. Jenny Emma Greene (CoI)              | Princeton University                                |

### OBSERVATIONS

| <i>Folder</i>      | <i>Observation</i> | <i>Label</i>            | <i>Observing Template</i>               | <i>Science Target</i>         |
|--------------------|--------------------|-------------------------|-----------------------------------------|-------------------------------|
| Observation Folder |                    |                         |                                         |                               |
|                    | 1                  | Wide-epoch1 WFSS-1      | NIRCam Wide Field Slitless Spectroscopy | (1) NEXUS-Center              |
|                    | 2                  | Deep-epoch1 NIRCam      | NIRCam Imaging                          | (1) NEXUS-Center              |
|                    | 3                  | Deep-epoch1 NIRSpec MOS | NIRSpec MultiObject Spectroscopy        | (3) nexus_msa_src_cat_updated |
|                    | 4                  | Deep-epoch2 NIRCam      | NIRCam Imaging                          | (1) NEXUS-Center              |
|                    | 5                  | Deep-epoch2 NIRSpec MOS | NIRSpec MultiObject Spectroscopy        | (4) msa_catalog_deep_ep2      |
|                    | 6                  | Deep-epoch3 NIRCam      | NIRCam Imaging                          | (1) NEXUS-Center              |

| <b>Folder</b> | <b>Observation</b> | <b>Label</b>                      | <b>Observing Template</b>               | <b>Science Target</b>             |
|---------------|--------------------|-----------------------------------|-----------------------------------------|-----------------------------------|
|               | 7                  | Deep-epoch3 NIRSpec MOS           | NIRSpec MultiObject Spectroscopy        | (7) msa_catalog_deep_ep3_enlarged |
|               | 8                  | Deep-epoch4 NIRCам                | NIRCам Imaging                          | (1) NEXUS-Center                  |
|               | 9                  | Deep-epoch4 NIRSpec MOS           | NIRSpec MultiObject Spectroscopy        | (8) msa_catalog_deep_ep4          |
|               | 10                 | Deep-epoch5 NIRCам                | NIRCам Imaging                          | (1) NEXUS-Center                  |
|               | 11                 | Deep-epoch5 NIRSpec MOS           | NIRSpec MultiObject Spectroscopy        | (9) msa_catalog_deep_ep5          |
|               | 12                 | Deep-epoch6 NIRCам                | NIRCам Imaging                          | (1) NEXUS-Center                  |
|               | 13                 | Deep-epoch6 NIRSpec MOS           | NIRSpec MultiObject Spectroscopy        | (10) msa_catalog_deep_ep6         |
|               | 14                 | Wide-epoch1 WFSS-2                | NIRCам Wide Field Slitless Spectroscopy | (1) NEXUS-Center                  |
|               | 15                 | Wide-epoch1 WFSS-2 Copy of Tile-6 | NIRCам Wide Field Slitless Spectroscopy | (5) NEXUS-Center-Tile-6           |

## ABSTRACT

NEXUS is a proposed multi-cycle treasury spectroscopic survey around the North Ecliptic Pole with exquisite optical through MIR photometry and temporal sampling. It contains two overlapping tiers. The Wide tier ( $\sim 400$  arcmin<sup>2</sup>) performs NIRCам/WFSS 2.4-5micron grism spectroscopy with three epochs over 3 cycles. The Deep tier ( $\sim 50$  arcmin<sup>2</sup>) performs high-multiplexing NIRSpec 0.6-5.3micron MOS/PRISM spectroscopy for up to  $\sim 10,000$  targets, over 18 epochs with a 2-month cadence. All epochs have simultaneous deep NIRCам and MIRI imaging. The field is within the continuous viewing zone of JWST, and is fully covered in the Euclid Deep Field North, with a wealth of deep multi-wavelength data to maximize synergy across wavelengths and science areas.

NEXUS has three science pillars. First, with its massive spectroscopic sample and deep photometry, it will perform efficient classification and physical characterization of galaxies and AGNs from  $z \sim 1$  to Cosmic Dawn. With the large contiguous area coverage, it will measure the spatial clustering and demography of the first galaxies and SMBHs at  $z > 6$ . Second, multi-epoch observations enable systematic time-domain investigations, focusing on  $z > 3$  supernova searches and low-mass AGN reverberation mapping. Third, the comprehensive data set from three JWST instruments will enable knowledge transfer to other legacy fields, create data challenges, and initiate benchmark work for future space missions. With rapid public releases of processed data and an open invitation for participation, NEXUS aims for broad and swift community engagement, to become a powerhouse to drive transformative advancements in multiple key science areas of astronomy.

## **OBSERVING DESCRIPTION**

The NEXUS project is a multi-instrument, multi-cycle (Cycles 3,4,5), multi-epoch JWST treasure survey, aiming at addressing three fundamental science thrusts: (1) Galaxy science with a focus on the  $z \sim 6$  populations; (2) Time-domain science with focuses on transients and AGN reverberation mapping; (3) Data challenges and applications to broader JWST data and astronomical communities. The program consists of two tiers of area coverage. A Wide tier covers  $\sim 400 \text{ arcmin}^2$  with NIRCam/WFSS grism spectroscopy (2.4-5 micron), and a Deep tier covers the central  $\sim 50 \text{ arcmin}^2$  with NIRSpec MOS/PRISM spectroscopy (0.6-5.3 micron). Both tiers have deep NIRCam imaging and parallel MIRI imaging to cover  $\sim 0.9$ -15 micron.

The time request is  $\sim$ equally divided in three cycles, with Cycle 3 receiving slightly more time. In Cycle 3, we request one Wide epoch and 6 Deep epochs. The total time charge in Cycle 3 is 129 hrs. Given the duration of Cycle 3 (July 1, 2024 – June 30, 2025), the Wide epoch should be scheduled in July (early Aug is OK) before the first Deep epoch. The Deep epochs have a nominal cadence of  $\sim 2$  months, though  $\pm 2$  weeks is also acceptable. [Currently in APT the Deep epochs are restricted to  $\pm 1$  week, which can be adjusted later. ]

Each Wide epoch will cost  $\sim 60$  hrs and each Deep epoch will cost  $\sim 10$  hrs.

The detailed observation designs and instrument configurations are described as follows:

\*\*\* Wide Tier:

### **1. NIRCam/WFSS**

We will use a combination of F322W2 and F444W filters in the LW channel of WFSS observations paired with F115W and F200W imaging in SW channel to cover a complete wavelength range from  $\sim 2.4$  to 5 micron with grism spectroscopy. We will also obtain F150W+F356W (in-field) and F090W+F444W (both in-field and out-field) direct imaging after WFSS exposures. We only use Grism-R for WFSS observations since the confusion will be resolved by the detection of multiple emission lines and three different PAs provided by three epochs.

We will adopt a 4 columns x 9 rows mosaic pattern to cover an area of  $21.4' \times 18.4' = 394 \text{ arcmin}^2$ . We will use a two-point INTRAMODULEX primary dither pattern to fill the gaps between SW detectors and mitigate the effect of cosmic rays. To save overhead, we do not require subpixel dither, as we will have three observations with different PAs. We will use the SHALLOW4 readout pattern with 6 Groups and 1 Intergration for

## JWST Proposal 5105 (Created: Tuesday, February 24, 2026, 3:01:50PM Eastern Standard Time) - Overview

F322W2 WFSS exposure and the MEDIUM8 with 6 Groups and 1 Intergration for F444W WFSS exposure. Both direct imaging exposures will be obtained with the SHALLOW4 readout pattern with 6 Groups and 1 Intergration. The above configurations will result in a total exposure time of 623s for F322W2 WFSS observations, 1245s for F444W WFSS observations, 311s for F150W+F356W direct imaging, and 934s for F090W+F444W direct imaging (in-field + out-field). Note these numbers are for a single epoch.

The Wide tier will have three repeated observations (three epochs) with PA rotations of 135 and 270 degrees from the first epoch. This arrangement will result in a cadence of ~16.5 month. Three different PAs will mitigate source confusion in the grism wavelength dispersion direction, and help fill the gaps between two NIRCam Modules. For epochs 2 and 3 in Cycles 4 and 5 respectively, we only require one set of direct imaging, with F150W+356W for epoch 2 and F090W+F444W for epoch 3. We choose the PA of epoch 1, which is scheduable during the first month of Cycle 3. With this fixed PA, we adjust the field center to avoid any bright stars ( $K_s < 10$  Vega mag) entering the FoV of our NIRCam observations. Combining three epochs, the coadded exposure is 1869s for F322W2 WFSS grism, and 3735s for F444W WFSS grism.

### 2. MIRI parallel

We will obtain coordinated MIRI parallel imaging in F770W+F1000W+F1280W. These MIRI imaging is important for source classification, target selection and detailed science studies. The F770W imaging will be obtained simultaneously in parallel with the F322W2 WFSS exposures with a total exposure time of 638s (957s in epoch 1 with an additional direct imaging exposure). F1000W will be obtained in parallel simultaneously with the F444W WFSS exposures with a total exposure time of 1260s. F1280W will be obtained simultaneously in parallel with the final direct imaging with a total exposure time of 957s. We will adopt the default FASTR1 readout pattern with 2 or 4 Integrations to avoid longer than 300s integration time.

### \*\*\* Deep Tier:

#### 1. NIRCam imaging primary + MIRI parallel

NIRCam primary imaging in the Deep tier is necessary to provide the cadenced imaging for time-domain studies, since the FoV coverages of NIRCam and NIRSpec are different (so we cannot use NIRCam parallel with NIRSpec primary). We will use F200W+F444W NIRCam imaging with a FULLBOX 2TIGHTGAPS primary dither to fill the gap between two NIRCam Modules. F200W is essential for high-redshift transient searches. The observations consist of 2 columns x 3 rows mosaics to cover the central  $10.8' \times 6.5' = 70 \text{ arcmin}^2$  of the Wide Tier. We will adopt the SHALLOW4 readout pattern with 6 Groups and 1 Integration (same as Wide Tier NIRcam direct imaging), with a total exposure time of 623s.

We will obtain coordinated MIRI parallel imaging with the NIRCam imaging primary in F1000W with FASTR1 readout pattern, 57 Groups, and 2 Integrations with a total exposure time of 638s. This is to provide additional time-resolved MIRI imaging in Deep to aid time-domain studies.

## 2. NIRSpec MOS primary + NIRCam imaging parallel

We will use NIRSpec MOS spectroscopy with a 2x2 mosaic to obtain deep prism ( $R=30-300$ ) 0.6-5.3 micron in the central  $\sim 7' \times 7' = 49 \text{ arcmin}^2$  covered by Deep Tier NIRCam imaging. We will adopt PRISM/CLEAR grating, NRSIRS2RAPID readout pattern, 43 Groups, and 1 Integration for each pointing. We will adopt a two shutter slitlet nodding to mitigate detector defects, help remove cosmic rays, improve flux accuracy, and improve spatial and spectral sampling. The total exposure time is 1284s per pointing.

We will obtain coordinated NIRCam parallel F210M+F360M imaging simultaneously with NIRSpec MOS observations. We will use the MEDIUM8 readout pattern with 6 Groups and 1 Intergrations. NIRCam parallel imaging adds two medium filters over  $\sim 6' \times 39' = 234 \text{ arcmin}^2$  fully within the Wide Tier NIRCam imaging to improve photometric classification therein. This filter combination can effectively select objects at  $z=4.6$  with both [O II]3727 and Halpha, [O III] emitters at  $z\sim 6.2$ , Hbeta emitters at  $z\sim 6.4$ , and [O II] emitters at  $z\sim 8.7$ .

To achieve the proposed science goals, we will repeat the above observations 18 times (epochs) spread over 3 years (cycles 3,4,5) with a cadence of  $\sim 2$  months and 6 PAs per year (rotating by  $\sim 60$  degrees). The total coadded exposure is  $623 \times 18 = 11214\text{s}$  (3.1h) for NIRCam imaging primary and  $1284 \times 18 = 23112\text{s}$  (6.4h) for NIRSpec MOS spectroscopy over 18 epochs.

Pointing positions of NIRSpec MOS observations are provisional, which would be updated after the first Wide Tier epoch.

Epoch 1 of Wide Tier and epochs 1-6 of Deep Tier observations can be scheduled in Cycle 3 with a total charged time of  $\sim 129\text{h}$ . Epoch 2 of Wide Tier and epochs 7-12 of Deep Tier observations will be scheduled in Cycle 4 (estimated total charged time of  $\sim 119\text{h}$ ), while Epoch 3 of Wide Tier and epochs 13-18 of Deep Tier observations will be scheduled in Cycle 5 (estimated total charged time of  $\sim 119\text{h}$ ).

# Proposal 5105 - Targets - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Fixed Targets | #                                                                       | Name                          | Target Coordinates                                                                  | Targ. Coord. Corrections | Miscellaneous |
|---------------|-------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------|--------------------------|---------------|
|               | (1)                                                                     | NEXUS-Center                  | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field] |                               |                                                                                     |                          |               |
|               | (3)                                                                     | nexus_msa_src_cat_updated     | RA: 17 53 50.4213 (268.4600887d)<br>Dec: +65 11 54.04 (65.19834d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Description=[]                                             |                               |                                                                                     |                          |               |
|               | (4)                                                                     | msa_catalog_deep_ep2          | RA: 17 53 50.5729 (268.4607204d)<br>Dec: +65 11 55.32 (65.19870d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Description=[]                                             |                               |                                                                                     |                          |               |
|               | (5)                                                                     | NEXUS-Center-Tile-6           | RA: 17 55 6.7769 (268.7782371d)<br>Dec: +65 10 43.06 (65.17863d)<br>Equinox: J2000  |                          |               |
|               | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field] |                               |                                                                                     |                          |               |
|               | (7)                                                                     | msa_catalog_deep_ep3_enlarged | RA: 17 53 50.4319 (268.4601329d)<br>Dec: +65 11 54.74 (65.19854d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Description=[]                                             |                               |                                                                                     |                          |               |
|               | (8)                                                                     | msa_catalog_deep_ep4          | RA: 17 53 50.4401 (268.4601671d)<br>Dec: +65 11 54.74 (65.19854d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Description=[]                                             |                               |                                                                                     |                          |               |
|               | (9)                                                                     | msa_catalog_deep_ep5          | RA: 17 53 50.4509 (268.4602121d)<br>Dec: +65 11 54.74 (65.19854d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Description=[]                                             |                               |                                                                                     |                          |               |
|               | (10)                                                                    | msa_catalog_deep_ep6          | RA: 17 53 50.4528 (268.4602200d)<br>Dec: +65 11 54.72 (65.19853d)<br>Equinox: J2000 |                          |               |
|               | Comments:<br>Description=[]                                             |                               |                                                                                     |                          |               |

# Proposal 5105 - Observation 1 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                                                                                          |                                                                                                                                                                                                                                                |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------|-------------------------------------------------------------------------------------|---------------|------------------|--------------------------|---------------------|-----------------|-------------------------|--------------------|---------------|------------------------------|--|
| Observation                                                                              | Proposal 5105, Observation 1: Wide-epoch1 WFSS-1                                                                                                                                                                                               |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               | Tue Feb 24 20:01:50 GMT 2026 |  |
|                                                                                          | Diagnostic Status: Warning                                                                                                                                                                                                                     |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | Observing Template: NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                                 |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
| Diagnostics                                                                              | (Wide-epoch1 WFSS-1 (Obs 1)) Warning (Form): This observation is split across multiple visits using multiple filters. Not selecting the sequence option may result in execution of the visits in a non-numerical order and is not recommended. |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Wide-epoch1 WFSS-1 (Obs 1)) Warning (Form): Use of only one of GRISMR or GRISMC may result in spectral overlap from multiple sources that can't be corrected. Users should address this issue in their proposal text.                         |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                    |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | (Visit 1:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                                   |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
| (Wide-epoch1 WFSS-1 (Obs 1)) Informational (Form): Target Placement is a required field. |                                                                                                                                                                                                                                                |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
| Fixed Targets                                                                            | #                                                                                                                                                                                                                                              | Name                |              | Target Coordinates                                                                  |               |                  | Targ. Coord. Corrections |                     |                 | Miscellaneous           |                    |               |                              |  |
|                                                                                          | (1)                                                                                                                                                                                                                                            | NEXUS-Center        |              | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                                        |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
| Template                                                                                 | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                                                        |                     |              |                                                                                     |               |                  | MIRI Imaging             |                     |                 |                         |                    |               |                              |  |
|                                                                                          | Module: ALL                                                                                                                                                                                                                                    |                     |              |                                                                                     |               |                  | Subarray: FULL           |                     |                 |                         |                    |               |                              |  |
|                                                                                          | Subarray: FULL                                                                                                                                                                                                                                 |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | Grism (Long Wavelength): GRISMR                                                                                                                                                                                                                |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
|                                                                                          | Show partial spectra region in Aladin: false                                                                                                                                                                                                   |                     |              |                                                                                     |               |                  |                          |                     |                 |                         |                    |               |                              |  |
| Mosaic                                                                                   | Rows                                                                                                                                                                                                                                           |                     | Columns      |                                                                                     | Row Overlap % |                  | Column Overlap %         |                     | Row shift (deg) |                         | Column shift (deg) |               | Tile Order                   |  |
|                                                                                          | 9                                                                                                                                                                                                                                              |                     | 4            |                                                                                     | 5.0           |                  | 5.0                      |                     | 0.0             |                         | 0.0                |               | HILBERT_CURVE                |  |
| Dithers                                                                                  | #                                                                                                                                                                                                                                              | Primary Dither Type |              |                                                                                     |               | Primary Dithers  |                          |                     |                 | Subpixel Positions      |                    |               |                              |  |
|                                                                                          | 1                                                                                                                                                                                                                                              |                     | INTRAMODULEX |                                                                                     |               |                  | 2                        |                     |                 |                         | NONE               |               |                              |  |
| Direct Image                                                                             | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                                                        | Short Filter        | Long Filter  | Readout Pattern                                                                     | Groups/Int    | Integrations/Exp | Total Integrations       | Total Exposure Time | Optional ETC ID | Grism (Long Wavelength) | Exposure Type      | Total Dithers |                              |  |
|                                                                                          | 1                                                                                                                                                                                                                                              | F150W               | F356W        | SHALLOW4                                                                            | 6             | 1                | 1                        | 311.366             |                 | GRISMR                  | Direct Image       | 1             |                              |  |
|                                                                                          | 2                                                                                                                                                                                                                                              | F090W               | F444W        | SHALLOW4                                                                            | 6             | 1                | 1                        | 311.366             |                 | GRISMR                  | Direct Image       | 1             |                              |  |

# Proposal 5105 - Observation 1 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                               | Short Filter | Long Filter     | Readout Pattern | Groups/Int       | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID    | Grism (Long Wavelength) | Exposure Type           | Total Dithers |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-----------------|-----------------|------------------|------------------|--------------------|---------------------|--------------------|-------------------------|-------------------------|---------------|
|                      | 1                                                                                                                                                                                                     | F115W        | F322W2          | SHALLOW4        | 6                | 1                | 2                  | 622.733             |                    | GRISMR                  | Grism (Long Wavelength) | 2             |
|                      | 2                                                                                                                                                                                                     | F200W        | F444W           | MEDIUM8         | 6                | 1                | 2                  | 1245.465            |                    | GRISMR                  | Grism (Long Wavelength) | 2             |
|                      | 3                                                                                                                                                                                                     | F090W        | F444W           | SHALLOW4        | 6                | 1                | 2                  | 622.733             |                    |                         | Out of Field            | 2             |
| Spectral Elements    | MIRI Imaging                                                                                                                                                                                          | Filter       | Readout Pattern | Groups/Int      | Integrations/Exp | Exposures/Dith   | Dither             | Total Dithers       | Total Integrations | Total Exposure Time     | Optional ETC ID         |               |
|                      | 1                                                                                                                                                                                                     | F770W        | FASTR1          | 57              | 2                | 1                |                    | 2                   | 4                  | 638.259                 |                         |               |
|                      | 2                                                                                                                                                                                                     | F770W        | FASTR1          | 57              | 2                | 1                |                    | 1                   | 2                  | 319.13                  |                         |               |
|                      | 3                                                                                                                                                                                                     | F1000W       | FASTR1          | 56              | 4                | 1                |                    | 2                   | 8                  | 1259.868                |                         |               |
|                      | 4                                                                                                                                                                                                     | F1280W       | FASTR1          | 57              | 2                | 1                |                    | 1                   | 2                  | 319.13                  |                         |               |
|                      | 5                                                                                                                                                                                                     | F1280W       | FASTR1          | 57              | 2                | 1                |                    | 2                   | 4                  | 638.259                 |                         |               |
| Special Requirements | Between Dates 25-AUG-2024:00:00:00 and 21-SEP-2024:00:00:00<br>Sequence Visits within 5 Days<br>Aperture PA Range 97.0 to 97.0 Degrees (V3 97.0 to 97.0)<br>Visits Same PA<br>No Parallel Attachments |              |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |
|                      | 3 After 1 by 60.0 Days to <None specified>                                                                                                                                                            |              |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |



# Proposal 5105 - Observation 2 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                   |                                                                                                                                                                                                                                       |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|-------------------------------------------------------------------------------------|------------|--------------------------|--------------------|--------------------|---------------------|----------------------------------------|------------------------------|-----------------------------|
| Observation       | Proposal 5105, Observation 2: Deep-epoch1 NIRCam                                                                                                                                                                                      |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        | Tue Feb 24 20:01:50 GMT 2026 |                             |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                            |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | Observing Template: NIRCam Imaging                                                                                                                                                                                                    |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                        |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
| Diagnostics       | (Deep-epoch1 NIRCam (Obs 2)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | (Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | (Visit 2:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | (Visit 2:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | (Visit 2:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | (Visit 2:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | (Visit 2:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
| Fixed Targets     | #                                                                                                                                                                                                                                     | Name                |             | Target Coordinates                                                                  |            | Targ. Coord. Corrections |                    |                    | Miscellaneous       |                                        |                              |                             |
|                   | (1)                                                                                                                                                                                                                                   | NEXUS-Center        |             | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                               |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
| Template          | NIRCam Imaging                                                                                                                                                                                                                        |                     |             |                                                                                     |            | MIRI Imaging             |                    |                    |                     |                                        |                              |                             |
|                   | Module: ALL                                                                                                                                                                                                                           |                     |             |                                                                                     |            | Subarray: FULL           |                    |                    |                     |                                        |                              |                             |
|                   | Subarray: FULL                                                                                                                                                                                                                        |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
|                   | Target Placement: Module gap (large extended source)                                                                                                                                                                                  |                     |             |                                                                                     |            |                          |                    |                    |                     |                                        |                              |                             |
| Mosaic            | Rows                                                                                                                                                                                                                                  | Columns             |             | Row Overlap %                                                                       |            | Column Overlap %         |                    | Row shift (deg)    |                     | Column shift (deg)                     |                              | Tile Order                  |
|                   | 3                                                                                                                                                                                                                                     | 2                   |             | 3.0                                                                                 |            | 3.0                      |                    | 0.0                |                     | 0.0                                    |                              | DEFAULT                     |
| Dithers           | #                                                                                                                                                                                                                                     | Primary Dither Type |             | Primary Dithers                                                                     |            | Dither Size              |                    | Subpixel Positions |                     | Coordinated Parallel Subpixel Selector |                              | Dither Direct Images Primes |
|                   | 1                                                                                                                                                                                                                                     | FULLBOX             |             | 2TIGHTGAPS                                                                          |            |                          |                    | 1                  |                     | NIRCam Only                            |                              | NO_DITHERING                |
| Spectral Elements | NIRCam Imaging                                                                                                                                                                                                                        | Short Filter        | Long Filter | Readout Pattern                                                                     | Groups/Int | Integrations/Exp         | Total Integrations | Total Dithers      | Total Exposure Time | Optional ETC ID                        |                              |                             |
|                   | 1                                                                                                                                                                                                                                     | F200W               | F444W       | SHALLOW4                                                                            | 6          | 1                        | 2                  | 2                  | 622.733             |                                        |                              |                             |

Proposal 5105 - Observation 2 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | MIRI Imaging                                                                                                                                                                                                                                                           | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|------------|------------------|----------------|--------|---------------|--------------------|---------------------|-----------------|
|                      | 1                                                                                                                                                                                                                                                                      | F1000W | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
| Special Requirements | Between Dates 30-MAY-2025:00:00:00 and 06-JUN-2025:00:00:00<br>Group Visits within 3 Days<br>Aperture PA Range 194.72542306 to 194.72542306 Degrees (V3 194.8 to 194.8)<br>Visits Same PA<br>No Parallel Attachments<br><br>Group Observations 2, 3, Non-interruptible |        |                 |            |                  |                |        |               |                    |                     |                 |

# Proposal 5105 - Observation 3 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|               |                                                                                             |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|---------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------|--------|---------------------|--------------------------|------------|------------------|--------------------|---------------------|-----------------|------------------------------|
| Observation   | Proposal 5105, Observation 3: Deep-epoch1 NIRSpec MOS                                       |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 | Tue Feb 24 20:01:50 GMT 2026 |
|               | Diagnostic Status: Warning                                                                  |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Observing Template: NIRSpec MultiObject Spectroscopy                                        |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Coordinated Parallel Template(s): NIRCcam Imaging                                           |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
| Diagnostics   | (Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 3:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 3:3) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 3:4) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
| Fixed Targets | #                                                                                           | Name                                                                              | Target Coordinates               |        |                     | Targ. Coord. Corrections |            |                  | Miscellaneous      |                     |                 |                              |
|               | (3)                                                                                         | nexus_msa_src_cat_updated                                                         | RA: 17 53 50.4213 (268.4600887d) |        |                     |                          |            |                  |                    |                     |                 |                              |
|               |                                                                                             |                                                                                   | Dec: +65 11 54.04 (65.19834d)    |        |                     |                          |            |                  |                    |                     |                 |                              |
|               |                                                                                             |                                                                                   | Equinox: J2000                   |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Comments:<br>Description=[]                                                                 |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
| Acquisition   | NIRSpec MultiObject Spectroscopy                                                            | Reference Star Bin                                                                | Target                           | Filter | MSA Configuration   | Readout Pattern          | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |                              |
|               | 1                                                                                           | Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                             | F140X  | Auto Acq MSA Config | NRSRAPID                 | 3          | 1                | 4                  | 171.788             |                 |                              |
|               | 2                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                             | F140X  | Auto Acq MSA Config | NRSRAPIDD6               | 3          | 1                | 4                  | 687.153             |                 |                              |
|               | 3                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                             | F140X  | Auto Acq MSA Config | NRSRAPIDD6               | 3          | 1                | 4                  | 687.153             |                 |                              |
|               | 4                                                                                           | Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                             | F140X  | Auto Acq MSA Config | NRSRAPID                 | 3          | 1                | 4                  | 171.788             |                 |                              |

# Proposal 5105 - Observation 3 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                 |                                                       |       |            |           |           |                |       |            |           |           |
|-----------------|-------------------------------------------------------|-------|------------|-----------|-----------|----------------|-------|------------|-----------|-----------|
| Template        | NIRSpec MultiObject Spectroscopy                      |       |            |           |           | NIRCam Imaging |       |            |           |           |
|                 | TA Method: MSATA                                      |       |            |           |           | Module: ALL    |       |            |           |           |
|                 | HFF Readout Mode: false                               |       |            |           |           | Subarray: FULL |       |            |           |           |
|                 | Obtain Confirmation Images: No                        |       |            |           |           |                |       |            |           |           |
|                 | Science Aperture: MSA Center                          |       |            |           |           |                |       |            |           |           |
|                 | Primary Candidate List: Epoch1 priority (317 sources) |       |            |           |           |                |       |            |           |           |
|                 | Filler Candidate List: Epoch1 filler (9204 sources)   |       |            |           |           |                |       |            |           |           |
| Reference Stars | Spectral Overlap Map: jwst-nirspec-prism              |       |            |           |           |                |       |            |           |           |
|                 | Spectral Overlap Threshold: 1.5                       |       |            |           |           |                |       |            |           |           |
|                 | Visit                                                 | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 1                                                     | 14377 | 268.355827 | 65.183088 | 21.86     | 1              | 28735 | 268.382658 | 65.233052 | 20.73     |
|                 | 1                                                     | 15647 | 268.371337 | 65.188058 | 21.70     | 1              | 31476 | 268.339780 | 65.219572 | 21.13     |
|                 | 1                                                     | 28046 | 268.394031 | 65.234944 | 21.66     | 1              | 31686 | 268.339565 | 65.218885 | 20.95     |
|                 | 1                                                     | 28199 | 268.400677 | 65.235957 | 21.09     | 1              | 35756 | 268.422131 | 65.197596 | 21.02     |
|                 | Visit                                                 | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 2                                                     | 18863 | 268.423306 | 65.255369 | 24.23     | 2              | 28964 | 268.526738 | 65.231962 | 23.26     |
|                 | 2                                                     | 20654 | 268.483254 | 65.258691 | 23.85     | 2              | 30605 | 268.451335 | 65.224157 | 23.59     |
|                 | 2                                                     | 21981 | 268.530983 | 65.267793 | 23.97     | 2              | 31376 | 268.477817 | 65.219947 | 24.99     |
|                 | 2                                                     | 27940 | 268.436336 | 65.237803 | 24.19     | 2              | 33200 | 268.463174 | 65.212309 | 22.78     |
|                 | Visit                                                 | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 3                                                     | 10528 | 268.563083 | 65.165806 | 23.10     | 3              | 19366 | 268.487633 | 65.196765 | 22.38     |
|                 | 3                                                     | 10888 | 268.611089 | 65.167682 | 23.79     | 3              | 33331 | 268.541756 | 65.210570 | 23.85     |
|                 | 3                                                     | 11347 | 268.557216 | 65.170067 | 24.23     | 3              | 34448 | 268.585236 | 65.206879 | 23.82     |
|                 | 3                                                     | 13235 | 268.485216 | 65.177987 | 23.46     | 3              | 35555 | 268.502198 | 65.199513 | 24.68     |
|                 | Visit                                                 | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 4                                                     | 3616  | 268.451433 | 65.129459 | 21.78     | 4              | 7418  | 268.477463 | 65.147173 | 20.89     |
|                 | 4                                                     | 4490  | 268.443882 | 65.133482 | 21.48     | 4              | 8012  | 268.375673 | 65.150181 | 21.84     |
|                 | 4                                                     | 5751  | 268.495557 | 65.139174 | 22.21     | 4              | 8096  | 268.368413 | 65.150718 | 21.78     |
|                 | 4                                                     | 6940  | 268.478238 | 65.144756 | 21.75     | 4              | 11520 | 268.465120 | 65.171333 | 20.82     |
| Dithers         | #                                                     |       |            |           |           | Dither Type    |       |            |           |           |
|                 | 1                                                     |       |            |           |           | NONE           |       |            |           |           |

Proposal 5105 - Observation 3 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | NIRSpec<br>MultiObject<br>Spectroscopy                                                                                                                                                                                                                   | Exposure<br>Specification | MSA<br>Configuration | Nod Pattern       | Pointing                                                        | Aperture PA            | Dispersion Offset<br>(Shutters) | Cross-Dispersion<br>Offset (Shutters) | Total Dithers          | Total<br>Integrations | Total Exposure<br>Time |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|-------------------|-----------------------------------------------------------------|------------------------|---------------------------------|---------------------------------------|------------------------|-----------------------|------------------------|
|                      | 1                                                                                                                                                                                                                                                        | 1<br>(PRISM/CLEAR)        | c1 : EP1pt1          | 2 Shutter Slitlet | 268.37249191666<br>666 Degrees<br>65.215059722222<br>22 Degrees | 333.29512708903<br>21  |                                 |                                       | 2                      | 2                     | 1283.822               |
|                      | 2                                                                                                                                                                                                                                                        | 1<br>(PRISM/CLEAR)        | c1 : EP1pt2          | 2 Shutter Slitlet | 268.48597508333<br>336 Degrees<br>65.241288888888<br>89 Degrees | 333.39807317667<br>62  |                                 |                                       | 2                      | 2                     | 1283.822               |
|                      | 3                                                                                                                                                                                                                                                        | 1<br>(PRISM/CLEAR)        | c1 : EP1pt3          | 2 Shutter Slitlet | 268.55191425<br>Degrees<br>65.185882222222<br>23 Degrees        | 333.45783663620<br>04  |                                 |                                       | 2                      | 2                     | 1283.822               |
|                      | 4                                                                                                                                                                                                                                                        | 1<br>(PRISM/CLEAR)        | c1 : EP1pt4          | 2 Shutter Slitlet | 268.437138875<br>Degrees<br>65.156310833333<br>34 Degrees       | 333.35373652354<br>076 |                                 |                                       | 2                      | 2                     | 1283.822               |
| Spectral Elements    | NIRCam Imaging                                                                                                                                                                                                                                           | Short Filter              | Long Filter          | Readout Pattern   | Groups/Int                                                      | Integrations/Exp       | Total Integrations              | Total Dithers                         | Total Exposure<br>Time | Optional ETC ID       |                        |
|                      | 1                                                                                                                                                                                                                                                        | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
|                      | 2                                                                                                                                                                                                                                                        | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
|                      | 3                                                                                                                                                                                                                                                        | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
|                      | 4                                                                                                                                                                                                                                                        | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
| Special Requirements | Sequence Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br>MSA Scheduled Aperture PA 333.3746 to 333.3746 Degrees (V3 194.8 to 194.8)<br><br>3 After 1 by 60.0 Days to <None specified><br>Group Observations 2, 3, Non-interruptible |                           |                      |                   |                                                                 |                        |                                 |                                       |                        |                       |                        |

# Proposal 5105 - Observation 4 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                   |                                                                                                                                                                                                                                       |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|----------------------------------|-----------------|------------------|--------------------------|-----------------|---------------------|--------------------|----------------------------------------|------------|-----------------------------|
| Observation       | Proposal 5105, Observation 4: Deep-epoch2 NIRCam                                                                                                                                                                                      |                     |             |                                  |                 |                  |                          |                 |                     |                    | Tue Feb 24 20:01:50 GMT 2026           |            |                             |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                            |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Observing Template: NIRCam Imaging                                                                                                                                                                                                    |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                        |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Diagnostics       | (Deep-epoch2 NIRCam (Obs 4)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 4:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 4:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 4:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 4:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 4:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Fixed Targets     | #                                                                                                                                                                                                                                     | Name                |             | Target Coordinates               |                 |                  | Targ. Coord. Corrections |                 |                     | Miscellaneous      |                                        |            |                             |
|                   | (1)                                                                                                                                                                                                                                   | NEXUS-Center        |             | RA: 17 53 51.0000 (268.4625000d) |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   |                                                                                                                                                                                                                                       |                     |             | Dec: +65 11 57.00 (65.19917d)    |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   |                                                                                                                                                                                                                                       |                     |             | Equinox: J2000                   |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                               |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Template          | NIRCam Imaging                                                                                                                                                                                                                        |                     |             |                                  |                 | MIRI Imaging     |                          |                 |                     |                    |                                        |            |                             |
|                   | Module: ALL                                                                                                                                                                                                                           |                     |             |                                  |                 | Subarray: FULL   |                          |                 |                     |                    |                                        |            |                             |
|                   | Subarray: FULL                                                                                                                                                                                                                        |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Target Placement: Module gap (large extended source)                                                                                                                                                                                  |                     |             |                                  |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Mosaic            | Rows                                                                                                                                                                                                                                  | Columns             |             | Row Overlap %                    |                 | Column Overlap % |                          | Row shift (deg) |                     | Column shift (deg) |                                        | Tile Order |                             |
|                   | 3                                                                                                                                                                                                                                     | 2                   |             | 3.0                              |                 | 3.0              |                          | 0.0             |                     | 0.0                |                                        | DEFAULT    |                             |
| Dithers           | #                                                                                                                                                                                                                                     | Primary Dither Type |             |                                  | Primary Dithers |                  | Dither Size              |                 | Subpixel Positions  |                    | Coordinated Parallel Subpixel Selector |            | Dither Direct Images Primes |
|                   | 1                                                                                                                                                                                                                                     | FULLBOX             |             |                                  | 2TIGHTGAPS      |                  |                          |                 | 1                   |                    | NIRCam Only                            |            | NO_DITHERING                |
| Spectral Elements | NIRCam Imaging                                                                                                                                                                                                                        | Short Filter        | Long Filter | Readout Pattern                  | Groups/Int      | Integrations/Exp | Total Integrations       | Total Dithers   | Total Exposure Time | Optional ETC ID    |                                        |            |                             |
|                   | 1                                                                                                                                                                                                                                     | F200W               | F444W       | SHALLOW4                         | 6               | 1                | 2                        | 2               | 622.733             |                    |                                        |            |                             |

Proposal 5105 - Observation 4 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | MIRI Imaging                                                                                                                                                                             | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|------------|------------------|----------------|--------|---------------|--------------------|---------------------|-----------------|
|                      | 1                                                                                                                                                                                        | F1000W | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
| Special Requirements | Between Dates 29-JUL-2025:00:00:00 and 05-AUG-2025:00:00:00<br>Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br><br>Group Observations 4, 5, Non-interruptible |        |                 |            |                  |                |        |               |                    |                     |                 |

# Proposal 5105 - Observation 5 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|               |                                                                                             |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|---------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|---------------------|-----------------|--------------------------|------------------|--------------------|---------------------|-----------------|------------------------------|
| Observation   | Proposal 5105, Observation 5: Deep-epoch2 NIRSpec MOS                                       |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 | Tue Feb 24 20:01:50 GMT 2026 |
|               | Diagnostic Status: Warning                                                                  |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Observing Template: NIRSpec MultiObject Spectroscopy                                        |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Coordinated Parallel Template(s): NIRCam Imaging                                            |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Diagnostics   | (Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 5:1) Warning (Form): The recommended value is 8 Reference Stars for this template.   |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 5:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 5:3) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 5:4) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Fixed Targets | #                                                                                           | Name                                                                              | Target Coordinates                                                                  |        |                     |                 | Targ. Coord. Corrections |                  |                    | Miscellaneous       |                 |                              |
|               | (4)                                                                                         | msa_catalog_deep_ep2                                                              | RA: 17 53 50.5729 (268.4607204d)<br>Dec: +65 11 55.32 (65.19870d)<br>Equinox: J2000 |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Comments:<br>Description=[]                                                                 |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Acquisition   | NIRSpec MultiObject Spectroscopy                                                            | Reference Star Bin                                                                | Target                                                                              | Filter | MSA Configuration   | Readout Pattern | Groups/Int               | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |                              |
|               | 1                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 6 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6      | 3                        | 1                | 4                  | 687.153             |                 |                              |
|               | 2                                                                                           | Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [ Optimal TA Accuracy ]   | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPID        | 3                        | 1                | 4                  | 171.788             |                 |                              |
|               | 3                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6      | 3                        | 1                | 4                  | 687.153             |                 |                              |
|               | 4                                                                                           | Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [ Optimal TA Accuracy ]   | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPID        | 3                        | 1                | 4                  | 171.788             |                 |                              |



# Proposal 5105 - Observation 5 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                 |                                                                                                                                                                                                                                                                                                   |       |            |           |           |                               |       |            |           |           |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|-----------|-----------|-------------------------------|-------|------------|-----------|-----------|
| Template        | NIRSpec MultiObject Spectroscopy                                                                                                                                                                                                                                                                  |       |            |           |           | NIRCam Imaging                |       |            |           |           |
|                 | TA Method: MSATA<br>HFF Readout Mode: false<br>Obtain Confirmation Images: No<br>Science Aperture: MSA Center<br>Primary Candidate List: Primary (357 sources)<br>Filler Candidate List: Secondary (19056 sources)<br>Spectral Overlap Map: jwst-nirspec-prism<br>Spectral Overlap Threshold: 1.5 |       |            |           |           | Module: ALL<br>Subarray: FULL |       |            |           |           |
| Reference Stars | Visit                                                                                                                                                                                                                                                                                             | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 1                                                                                                                                                                                                                                                                                                 | 3748  | 268.334658 | 65.147695 | 24.93     | 1                             | 23017 | 268.359069 | 65.204644 | 24.63     |
|                 | 1                                                                                                                                                                                                                                                                                                 | 4201  | 268.415970 | 65.151296 | 24.55     | 1                             | 25522 | 268.414794 | 65.199150 | 24.37     |
|                 | 1                                                                                                                                                                                                                                                                                                 | 4803  | 268.335343 | 65.156606 | 24.58     | 1                             | 25810 | 268.414441 | 65.196213 | 23.66     |
|                 | Visit                                                                                                                                                                                                                                                                                             | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 2                                                                                                                                                                                                                                                                                                 | 14901 | 268.449140 | 65.261198 | 21.22     | 2                             | 21645 | 268.339789 | 65.219568 | 20.96     |
|                 | 2                                                                                                                                                                                                                                                                                                 | 20268 | 268.445925 | 65.226521 | 22.01     | 2                             | 21845 | 268.339570 | 65.218882 | 20.91     |
|                 | 2                                                                                                                                                                                                                                                                                                 | 20496 | 268.462523 | 65.225155 | 22.00     | 2                             | 23221 | 268.436240 | 65.213318 | 22.03     |
|                 | 2                                                                                                                                                                                                                                                                                                 | 21424 | 268.446359 | 65.220459 | 22.01     | 2                             | 24586 | 268.379736 | 65.206214 | 21.91     |
|                 | Visit                                                                                                                                                                                                                                                                                             | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 3                                                                                                                                                                                                                                                                                                 | 12818 | 268.487636 | 65.196766 | 22.34     | 3                             | 19770 | 268.494547 | 65.229004 | 23.87     |
|                 | 3                                                                                                                                                                                                                                                                                                 | 16774 | 268.514870 | 65.251235 | 22.85     | 3                             | 19975 | 268.578706 | 65.228553 | 23.58     |
|                 | 3                                                                                                                                                                                                                                                                                                 | 17022 | 268.510211 | 65.249256 | 23.34     | 3                             | 22686 | 268.588102 | 65.214919 | 23.12     |
|                 | 3                                                                                                                                                                                                                                                                                                 | 19295 | 268.526722 | 65.231959 | 23.17     | 3                             | 24455 | 268.585227 | 65.206878 | 23.79     |
|                 | Visit                                                                                                                                                                                                                                                                                             | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 4                                                                                                                                                                                                                                                                                                 | 3191  | 268.478235 | 65.144755 | 21.76     | 4                             | 5259  | 268.488334 | 65.161616 | 21.73     |
|                 | 4                                                                                                                                                                                                                                                                                                 | 3621  | 268.455551 | 65.147242 | 21.27     | 4                             | 8738  | 268.498638 | 65.179475 | 21.71     |
|                 | 4                                                                                                                                                                                                                                                                                                 | 3731  | 268.453241 | 65.148267 | 21.39     | 4                             | 11034 | 268.470433 | 65.189130 | 21.68     |
|                 | 4                                                                                                                                                                                                                                                                                                 | 4970  | 268.559472 | 65.159938 | 21.42     | 4                             | 12136 | 268.499538 | 65.194530 | 20.61     |
| Dithers         | #                                                                                                                                                                                                                                                                                                 |       |            |           |           | Dither Type                   |       |            |           |           |
|                 | 1                                                                                                                                                                                                                                                                                                 |       |            |           |           | NONE                          |       |            |           |           |

Proposal 5105 - Observation 5 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | NIRSpec MultiObject Spectroscopy                                                                                                                      | Exposure Specification | MSA Configuration | Nod Pattern       | Pointing                                               | Aperture PA         | Dispersion Offset (Shutters) | Cross-Dispersion Offset (Shutters) | Total Dithers       | Total Integrations | Total Exposure Time |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|-------------------|--------------------------------------------------------|---------------------|------------------------------|------------------------------------|---------------------|--------------------|---------------------|
|                      | 1                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : EP2PT1       | 2 Shutter Slitlet | 268.39177616666 666 Degrees 65.173404166666 67 Degrees | 276.51207015288 674 |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 2                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : EP2PT2       | 2 Shutter Slitlet | 268.408096375 Degrees 65.234194722222 23 Degrees       | 276.52677960738 094 |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 3                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : EP2PT3       | 2 Shutter Slitlet | 268.54390108333 33 Degrees 65.220861666666 66 Degrees  | 276.64999795360 07  |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 4                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : EP2PT4       | 2 Shutter Slitlet | 268.51742104166 664 Degrees 65.168141111111 11 Degrees | 276.62604082912 72  |                              |                                    | 2                   | 2                  | 1283.822            |
| Spectral Elements    | NIRCam Imaging                                                                                                                                        | Short Filter           | Long Filter       | Readout Pattern   | Groups/Int                                             | Integrations/Exp    | Total Integrations           | Total Dithers                      | Total Exposure Time | Optional ETC ID    |                     |
|                      | 1                                                                                                                                                     | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 2                                                                                                                                                     | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 3                                                                                                                                                     | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 4                                                                                                                                                     | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
| Special Requirements | Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br>MSA Scheduled Aperture PA 276.5746 to 276.5746 Degrees (V3 138.0 to 138.0) |                        |                   |                   |                                                        |                     |                              |                                    |                     |                    |                     |
|                      | Group Observations 4, 5, Non-interruptible                                                                                                            |                        |                   |                   |                                                        |                     |                              |                                    |                     |                    |                     |

# Proposal 5105 - Observation 6 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                   |                                                                                                                                                                                                                                       |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------|-----------------|------------------|------------------|--------------------------|--------------------|---------------------|----------------------------------------|------------------------------|
| Observation       | Proposal 5105, Observation 6: Deep-epoch3 NIRCam                                                                                                                                                                                      |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        | Tue Feb 24 20:01:50 GMT 2026 |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                            |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | Observing Template: NIRCam Imaging                                                                                                                                                                                                    |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                        |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
| Diagnostics       | (Deep-epoch3 NIRCam (Obs 6)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | (Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | (Visit 6:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | (Visit 6:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | (Visit 6:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | (Visit 6:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | (Visit 6:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
| Fixed Targets     | #                                                                                                                                                                                                                                     | Name                | Target Coordinates                                                                  |                 |                  |                  | Targ. Coord. Corrections |                    |                     | Miscellaneous                          |                              |
|                   | (1)                                                                                                                                                                                                                                   | NEXUS-Center        | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                               |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   |                                                                                                                                                                                                                                       |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
| Template          | NIRCam Imaging                                                                                                                                                                                                                        |                     |                                                                                     |                 |                  | MIRI Imaging     |                          |                    |                     |                                        |                              |
|                   | Module: ALL                                                                                                                                                                                                                           |                     |                                                                                     |                 |                  | Subarray: FULL   |                          |                    |                     |                                        |                              |
|                   | Subarray: FULL                                                                                                                                                                                                                        |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
|                   | Target Placement: Module gap (large extended source)                                                                                                                                                                                  |                     |                                                                                     |                 |                  |                  |                          |                    |                     |                                        |                              |
| Mosaic            | Rows                                                                                                                                                                                                                                  | Columns             | Row Overlap %                                                                       |                 | Column Overlap % |                  | Row shift (deg)          |                    | Column shift (deg)  |                                        | Tile Order                   |
|                   | 3                                                                                                                                                                                                                                     | 2                   | 3.0                                                                                 |                 | 3.0              |                  | 0.0                      |                    | 0.0                 |                                        | DEFAULT                      |
| Dithers           | #                                                                                                                                                                                                                                     | Primary Dither Type |                                                                                     | Primary Dithers |                  | Dither Size      |                          | Subpixel Positions |                     | Coordinated Parallel Subpixel Selector | Dither Direct Images Primes  |
|                   | 1                                                                                                                                                                                                                                     | FULLBOX             |                                                                                     | 2TIGHTGAPS      |                  |                  |                          | 1                  |                     | NIRCam Only                            | NO_DITHERING                 |
| Spectral Elements | NIRCam Imaging                                                                                                                                                                                                                        | Short Filter        | Long Filter                                                                         | Readout Pattern | Groups/Int       | Integrations/Exp | Total Integrations       | Total Dithers      | Total Exposure Time | Optional ETC ID                        |                              |
|                   | 1                                                                                                                                                                                                                                     | F200W               | F444W                                                                               | SHALLOW4        | 6                | 1                | 2                        | 2                  | 622.733             |                                        |                              |

Proposal 5105 - Observation 6 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | MIRI Imaging                                                                                                              | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|----------------------|---------------------------------------------------------------------------------------------------------------------------|--------|-----------------|------------|------------------|----------------|--------|---------------|--------------------|---------------------|-----------------|
|                      | 1                                                                                                                         | F1000W | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
| Special Requirements | Between Dates 27-SEP-2025:00:00:00 and 04-OCT-2025:00:00:00                                                               |        |                 |            |                  |                |        |               |                    |                     |                 |
|                      | Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br><br>Group Observations 6, 7, Non-interruptible |        |                 |            |                  |                |        |               |                    |                     |                 |

# Proposal 5105 - Observation 7 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|               |                                                                                             |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
|---------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|---------------------|--------------------------|------------|------------------|--------------------|---------------------|-----------------|------------------------------|
| Observation   | Proposal 5105, Observation 7: Deep-epoch3 NIRSpec MOS                                       |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 | Tue Feb 24 20:01:50 GMT 2026 |
|               | Diagnostic Status: Warning                                                                  |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Observing Template: NIRSpec MultiObject Spectroscopy                                        |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Coordinated Parallel Template(s): NIRCам Imaging                                            |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
| Diagnostics   | (Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 7:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 7:3) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 7:4) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
| Fixed Targets | #                                                                                           | Name                                                                              | Target Coordinates                                                                  |        |                     | Targ. Coord. Corrections |            |                  | Miscellaneous      |                     |                 |                              |
|               | (7)                                                                                         | msa_catalog_deep_ep3_enlarged                                                     | RA: 17 53 50.4319 (268.4601329d)<br>Dec: +65 11 54.74 (65.19854d)<br>Equinox: J2000 |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Comments:<br>Description=[]                                                                 |                                                                                   |                                                                                     |        |                     |                          |            |                  |                    |                     |                 |                              |
| Acquisition   | NIRSpec MultiObject Spectroscopy                                                            | Reference Star Bin                                                                | Target                                                                              | Filter | MSA Configuration   | Readout Pattern          | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |                              |
|               | 1                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6               | 3          | 1                | 4                  | 687.153             |                 |                              |
|               | 2                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6               | 3          | 1                | 4                  | 687.153             |                 |                              |
|               | 3                                                                                           | Filter: F110W; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                                                                                | F110W  | Auto Acq MSA Config | NRSRAPID                 | 3          | 1                | 4                  | 171.788             |                 |                              |
|               | 4                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6               | 3          | 1                | 4                  | 687.153             |                 |                              |

# Proposal 5105 - Observation 7 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                 |                                                                                                                                                                                                                                                                                                                             |       |            |           |           |                               |       |            |           |           |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------------|-----------|-----------|-------------------------------|-------|------------|-----------|-----------|
| Template        | NIRSpec MultiObject Spectroscopy                                                                                                                                                                                                                                                                                            |       |            |           |           | NIRCam Imaging                |       |            |           |           |
|                 | TA Method: MSATA<br>HFF Readout Mode: false<br>Obtain Confirmation Images: No<br>Science Aperture: MSA Center<br>Primary Candidate List: EP3 enlarged Primary (288 sources)<br>Filler Candidate List: EP3 enlarged Secondary (20808 sources)<br>Spectral Overlap Map: jwst-nirspec-prism<br>Spectral Overlap Threshold: 1.5 |       |            |           |           | Module: ALL<br>Subarray: FULL |       |            |           |           |
| Reference Stars | Visit                                                                                                                                                                                                                                                                                                                       | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 1                                                                                                                                                                                                                                                                                                                           | 13524 | 268.544884 | 65.273218 | 23.86     | 1                             | 20601 | 268.498474 | 65.224230 | 23.64     |
|                 | 1                                                                                                                                                                                                                                                                                                                           | 13625 | 268.548628 | 65.269437 | 23.77     | 1                             | 21358 | 268.567362 | 65.223695 | 22.65     |
|                 | 1                                                                                                                                                                                                                                                                                                                           | 14384 | 268.567272 | 65.266098 | 22.74     | 1                             | 22686 | 268.588094 | 65.214919 | 23.12     |
|                 | 1                                                                                                                                                                                                                                                                                                                           | 16475 | 268.592827 | 65.253418 | 22.59     | 1                             | 23396 | 268.541732 | 65.210567 | 23.78     |
|                 | Visit                                                                                                                                                                                                                                                                                                                       | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 2                                                                                                                                                                                                                                                                                                                           | 17871 | 268.376851 | 65.242201 | 24.86     | 2                             | 20847 | 268.451331 | 65.224156 | 23.59     |
|                 | 2                                                                                                                                                                                                                                                                                                                           | 18197 | 268.398638 | 65.238149 | 24.29     | 2                             | 23017 | 268.359063 | 65.204641 | 24.63     |
|                 | 2                                                                                                                                                                                                                                                                                                                           | 18315 | 268.375738 | 65.238361 | 23.74     | 2                             | 23273 | 268.463176 | 65.212310 | 22.78     |
|                 | 2                                                                                                                                                                                                                                                                                                                           | 18859 | 268.395525 | 65.233029 | 24.86     | 2                             | 25522 | 268.414789 | 65.199150 | 24.37     |
|                 | Visit                                                                                                                                                                                                                                                                                                                       | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 3                                                                                                                                                                                                                                                                                                                           | 1604  | 268.363024 | 65.135446 | 20.46     | 3                             | 4215  | 268.312874 | 65.152855 | 19.73     |
|                 | 3                                                                                                                                                                                                                                                                                                                           | 2457  | 268.355931 | 65.140968 | 19.56     | 3                             | 8384  | 268.400404 | 65.177881 | 20.88     |
|                 | 3                                                                                                                                                                                                                                                                                                                           | 3966  | 268.379478 | 65.150618 | 20.43     | 3                             | 9619  | 268.355829 | 65.183088 | 21.91     |
|                 | 3                                                                                                                                                                                                                                                                                                                           | 4016  | 268.375666 | 65.150181 | 21.92     | 3                             | 10827 | 268.371336 | 65.188058 | 21.62     |
|                 | Visit                                                                                                                                                                                                                                                                                                                       | ID    | RA         | Dec       | Magnitude | Visit                         | ID    | RA         | Dec       | Magnitude |
|                 | 4                                                                                                                                                                                                                                                                                                                           | 5146  | 268.524708 | 65.159957 | 23.29     | 4                             | 8906  | 268.480777 | 65.180494 | 24.28     |
|                 | 4                                                                                                                                                                                                                                                                                                                           | 7076  | 268.584832 | 65.171341 | 24.79     | 4                             | 17138 | 268.487671 | 65.202981 | 24.15     |
|                 | 4                                                                                                                                                                                                                                                                                                                           | 7607  | 268.493093 | 65.174030 | 23.27     | 4                             | 24872 | 268.525240 | 65.203798 | 23.54     |
|                 | 4                                                                                                                                                                                                                                                                                                                           | 8478  | 268.585578 | 65.177707 | 24.66     | 4                             | 25517 | 268.502186 | 65.199513 | 24.64     |
| Dithers         | #                                                                                                                                                                                                                                                                                                                           |       |            |           |           | Dither Type                   |       |            |           |           |
|                 | 1                                                                                                                                                                                                                                                                                                                           |       |            |           |           | NONE                          |       |            |           |           |

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| Spectral Elements    | NIRSpec MultiObject Spectroscopy                                                                                                                                                                      | Exposure Specification | MSA Configuration  | Nod Pattern       | Pointing                                               | Aperture PA         | Dispersion Offset (Shutters) | Cross-Dispersion Offset (Shutters) | Total Dithers       | Total Integrations | Total Exposure Time |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|-------------------|--------------------------------------------------------|---------------------|------------------------------|------------------------------------|---------------------|--------------------|---------------------|
|                      | 1                                                                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : ep3pt4_offset | 2 Shutter Slitlet | 268.54919741666 663 Degrees 65.239999166666 66 Degrees | 218.65534835061 95  |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 2                                                                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : ep3pt3_offset | 2 Shutter Slitlet | 268.41113166666 66 Degrees 65.218215277777 77 Degrees  | 218.53004008237 11  |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 3                                                                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : ep3pt2_offset | 2 Shutter Slitlet | 268.36655879166 665 Degrees 65.165780555555 56 Degrees | 218.48968028733 225 |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 4                                                                                                                                                                                                     | 1 (PRISM/CLEAR)        | c1 : ep3pt1_offset | 2 Shutter Slitlet | 268.52596558333 33 Degrees 65.176237777777 79 Degrees  | 218.63438596028 186 |                              |                                    | 2                   | 2                  | 1283.822            |
| Spectral Elements    | NIRCam Imaging                                                                                                                                                                                        | Short Filter           | Long Filter        | Readout Pattern   | Groups/Int                                             | Integrations/Exp    | Total Integrations           | Total Dithers                      | Total Exposure Time | Optional ETC ID    |                     |
|                      | 1                                                                                                                                                                                                     | F210M                  | F360M              | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 2                                                                                                                                                                                                     | F210M                  | F360M              | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 3                                                                                                                                                                                                     | F210M                  | F360M              | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 4                                                                                                                                                                                                     | F210M                  | F360M              | MEDIUM8           | 6                                                      | 1                   | 2                            | 2                                  | 1245.465            |                    |                     |
| Special Requirements | Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br>MSA Scheduled Aperture PA 218.5746 to 218.5746 Degrees (V3 80.0 to 80.0)<br><br>Group Observations 6, 7, Non-interruptible |                        |                    |                   |                                                        |                     |                              |                                    |                     |                    |                     |

# Proposal 5105 - Observation 8 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                   |                                                                                                                                                                                                                                       |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|-------------------------------------------------------------------------------------|-----------------|------------------|--------------------------|-----------------|---------------------|--------------------|----------------------------------------|------------|-----------------------------|
| Observation       | Proposal 5105, Observation 8: Deep-epoch4 NIRCam                                                                                                                                                                                      |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    | Tue Feb 24 20:01:50 GMT 2026           |            |                             |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                            |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Observing Template: NIRCam Imaging                                                                                                                                                                                                    |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                        |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Diagnostics       | (Deep-epoch4 NIRCam (Obs 8)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 8:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 8:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 8:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 8:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 8:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Fixed Targets     | #                                                                                                                                                                                                                                     | Name                |             | Target Coordinates                                                                  |                 |                  | Targ. Coord. Corrections |                 |                     | Miscellaneous      |                                        |            |                             |
|                   | (1)                                                                                                                                                                                                                                   | NEXUS-Center        |             | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                               |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Template          | NIRCam Imaging                                                                                                                                                                                                                        |                     |             |                                                                                     |                 | MIRI Imaging     |                          |                 |                     |                    |                                        |            |                             |
|                   | Module: ALL                                                                                                                                                                                                                           |                     |             |                                                                                     |                 | Subarray: FULL   |                          |                 |                     |                    |                                        |            |                             |
|                   | Subarray: FULL                                                                                                                                                                                                                        |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Target Placement: Module gap (large extended source)                                                                                                                                                                                  |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Mosaic            | Rows                                                                                                                                                                                                                                  | Columns             |             | Row Overlap %                                                                       |                 | Column Overlap % |                          | Row shift (deg) |                     | Column shift (deg) |                                        | Tile Order |                             |
|                   | 3                                                                                                                                                                                                                                     | 2                   |             | 3.0                                                                                 |                 | 3.0              |                          | 0.0             |                     | 0.0                |                                        | DEFAULT    |                             |
| Dithers           | #                                                                                                                                                                                                                                     | Primary Dither Type |             |                                                                                     | Primary Dithers |                  | Dither Size              |                 | Subpixel Positions  |                    | Coordinated Parallel Subpixel Selector |            | Dither Direct Images Primes |
|                   | 1                                                                                                                                                                                                                                     | FULLBOX             |             |                                                                                     | 2TIGHTGAPS      |                  |                          |                 | 1                   |                    | NIRCam Only                            |            | NO_DITHERING                |
| Spectral Elements | NIRCam Imaging                                                                                                                                                                                                                        | Short Filter        | Long Filter | Readout Pattern                                                                     | Groups/Int      | Integrations/Exp | Total Integrations       | Total Dithers   | Total Exposure Time | Optional ETC ID    |                                        |            |                             |
|                   | 1                                                                                                                                                                                                                                     | F200W               | F444W       | SHALLOW4                                                                            | 6               | 1                | 2                        | 2               | 622.733             |                    |                                        |            |                             |



Proposal 5105 - Observation 8 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | MIRI Imaging                                                                                                                                                                             | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|------------|------------------|----------------|--------|---------------|--------------------|---------------------|-----------------|
|                      | 1                                                                                                                                                                                        | F1000W | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
| Special Requirements | Between Dates 26-NOV-2025:00:00:00 and 03-DEC-2025:00:00:00<br>Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br><br>Group Observations 8, 9, Non-interruptible |        |                 |            |                  |                |        |               |                    |                     |                 |

# Proposal 5105 - Observation 9 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|               |                                                                                             |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|---------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------|--------|---------------------|--------------------------|------------|------------------|--------------------|---------------------|-----------------|------------------------------|
| Observation   | Proposal 5105, Observation 9: Deep-epoch4 NIRSpec MOS                                       |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 | Tue Feb 24 20:01:50 GMT 2026 |
|               | Diagnostic Status: Warning                                                                  |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Observing Template: NIRSpec MultiObject Spectroscopy                                        |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Coordinated Parallel Template(s): NIRCcam Imaging                                           |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
| Diagnostics   | (Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 9:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 9:3) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | (Visit 9:4) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
| Fixed Targets | #                                                                                           | Name                                                                              | Target Coordinates               |        |                     | Targ. Coord. Corrections |            |                  | Miscellaneous      |                     |                 |                              |
|               | (8)                                                                                         | msa_catalog_deep_ep4                                                              | RA: 17 53 50.4401 (268.4601671d) |        |                     |                          |            |                  |                    |                     |                 |                              |
|               |                                                                                             |                                                                                   | Dec: +65 11 54.74 (65.19854d)    |        |                     |                          |            |                  |                    |                     |                 |                              |
|               |                                                                                             |                                                                                   | Equinox: J2000                   |        |                     |                          |            |                  |                    |                     |                 |                              |
|               | Comments:<br>Description=[]                                                                 |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |
| Acquisition   | NIRSpec MultiObject Spectroscopy                                                            | Reference Star Bin                                                                | Target                           | Filter | MSA Configuration   | Readout Pattern          | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |                              |
|               | 1                                                                                           | Filter: F110W; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                             | F110W  | Auto Acq MSA Config | NRSRAPID                 | 3          | 1                | 4                  | 171.788             |                 |                              |
|               | 2                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                             | F140X  | Auto Acq MSA Config | NRSRAPIDD6               | 3          | 1                | 4                  | 687.153             |                 |                              |
|               | 3                                                                                           | Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                             | F140X  | Auto Acq MSA Config | NRSRAPID                 | 3          | 1                | 4                  | 171.788             |                 |                              |
|               | 4                                                                                           | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 3 quads; [ Optimal TA Accuracy ] | SAME                             | F140X  | Auto Acq MSA Config | NRSRAPIDD6               | 3          | 1                | 4                  | 687.153             |                 |                              |
|               |                                                                                             |                                                                                   |                                  |        |                     |                          |            |                  |                    |                     |                 |                              |

# Proposal 5105 - Observation 9 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                 |                                                  |       |            |           |           |                |       |            |           |           |
|-----------------|--------------------------------------------------|-------|------------|-----------|-----------|----------------|-------|------------|-----------|-----------|
| Template        | NIRSpec MultiObject Spectroscopy                 |       |            |           |           | NIRCam Imaging |       |            |           |           |
|                 | TA Method: MSATA                                 |       |            |           |           | Module: ALL    |       |            |           |           |
|                 | HFF Readout Mode: false                          |       |            |           |           | Subarray: FULL |       |            |           |           |
|                 | Obtain Confirmation Images: No                   |       |            |           |           |                |       |            |           |           |
|                 | Science Aperture: MSA Center                     |       |            |           |           |                |       |            |           |           |
|                 | Primary Candidate List: Primary (233 sources)    |       |            |           |           |                |       |            |           |           |
|                 | Filler Candidate List: Secondary (19979 sources) |       |            |           |           |                |       |            |           |           |
| Reference Stars | Spectral Overlap Map: jwst-nirspec-prism         |       |            |           |           |                |       |            |           |           |
|                 | Spectral Overlap Threshold: 1.5                  |       |            |           |           |                |       |            |           |           |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 1                                                | 5016  | 268.549348 | 65.159349 | 21.08     | 1              | 10930 | 268.510877 | 65.188622 | 20.98     |
|                 | 1                                                | 7785  | 268.620967 | 65.176840 | 19.73     | 1              | 12136 | 268.499536 | 65.194528 | 20.71     |
|                 | 1                                                | 9527  | 268.507806 | 65.183118 | 20.34     | 1              | 17402 | 268.567409 | 65.203702 | 20.04     |
|                 | 1                                                | 9589  | 268.486441 | 65.182937 | 20.88     | 1              | 23021 | 268.583510 | 65.213745 | 20.45     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 2                                                | 1620  | 268.414601 | 65.133458 | 24.91     | 2              | 4201  | 268.415961 | 65.151296 | 24.78     |
|                 | 2                                                | 2672  | 268.393062 | 65.141050 | 23.72     | 2              | 7038  | 268.428902 | 65.171184 | 24.69     |
|                 | 2                                                | 2695  | 268.390840 | 65.141033 | 24.63     | 2              | 7739  | 268.342806 | 65.174334 | 24.50     |
|                 | 2                                                | 3479  | 268.467042 | 65.146178 | 22.31     | 2              | 8567  | 268.387601 | 65.178115 | 24.31     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 3                                                | 10827 | 268.371336 | 65.188058 | 21.87     | 3              | 19146 | 268.294386 | 65.232448 | 21.11     |
|                 | 3                                                | 17753 | 268.300144 | 65.220699 | 20.62     | 3              | 23221 | 268.436243 | 65.213319 | 22.10     |
|                 | 3                                                | 18468 | 268.394017 | 65.234937 | 21.90     | 3              | 25466 | 268.311139 | 65.198993 | 22.19     |
|                 | 3                                                | 18730 | 268.398739 | 65.234018 | 22.23     | 3              | 25708 | 268.422140 | 65.197600 | 20.95     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 4                                                | 13625 | 268.548628 | 65.269437 | 24.05     | 4              | 19770 | 268.494548 | 65.229005 | 24.14     |
|                 | 4                                                | 15920 | 268.453697 | 65.256558 | 23.80     | 4              | 20601 | 268.498474 | 65.224230 | 23.56     |
|                 | 4                                                | 16774 | 268.514867 | 65.251232 | 23.05     | 4              | 22026 | 268.447368 | 65.217907 | 23.91     |
|                 | 4                                                | 18377 | 268.436340 | 65.237804 | 24.42     | 4              | 23506 | 268.503012 | 65.211410 | 24.61     |
| Dithers         | #                                                |       |            |           |           | Dither Type    |       |            |           |           |
|                 | 1                                                |       |            |           |           | NONE           |       |            |           |           |

Proposal 5105 - Observation 9 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | NIRSpec<br>MultiObject<br>Spectroscopy                                                                                                                                                                | Exposure<br>Specification | MSA<br>Configuration | Nod Pattern       | Pointing                                                        | Aperture PA            | Dispersion Offset<br>(Shutters) | Cross-Dispersion<br>Offset (Shutters) | Total Dithers          | Total<br>Integrations | Total Exposure<br>Time |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------|-------------------|-----------------------------------------------------------------|------------------------|---------------------------------|---------------------------------------|------------------------|-----------------------|------------------------|
|                      | 1                                                                                                                                                                                                     | 1<br>(PRISM/CLEAR)        | c1 : ep4pt1          | 2 Shutter Slitlet | 268.55423220833<br>33 Degrees<br>65.183225000000<br>01 Degrees  | 158.66004129666<br>19  |                                 |                                       | 2                      | 2                     | 1283.822               |
|                      | 2                                                                                                                                                                                                     | 1<br>(PRISM/CLEAR)        | c1 : ep4pt2          | 2 Shutter Slitlet | 268.42339304166<br>666 Degrees<br>65.160769722222<br>23 Degrees | 158.54118985269<br>255 |                                 |                                       | 2                      | 2                     | 1283.822               |
|                      | 3                                                                                                                                                                                                     | 1<br>(PRISM/CLEAR)        | c1 : ep4pt3          | 2 Shutter Slitlet | 268.37066074999<br>996 Degrees<br>65.215556388888<br>88 Degrees | 158.49322684126<br>58  |                                 |                                       | 2                      | 2                     | 1283.822               |
|                      | 4                                                                                                                                                                                                     | 1<br>(PRISM/CLEAR)        | c1 : ep4pt4          | 2 Shutter Slitlet | 268.503996125<br>Degrees<br>65.2379925<br>Degrees               | 158.61437187301<br>763 |                                 |                                       | 2                      | 2                     | 1283.822               |
| Spectral Elements    | NIRCam Imaging                                                                                                                                                                                        | Short Filter              | Long Filter          | Readout Pattern   | Groups/Int                                                      | Integrations/Exp       | Total Integrations              | Total Dithers                         | Total Exposure<br>Time | Optional ETC ID       |                        |
|                      | 1                                                                                                                                                                                                     | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
|                      | 2                                                                                                                                                                                                     | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
|                      | 3                                                                                                                                                                                                     | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
|                      | 4                                                                                                                                                                                                     | F210M                     | F360M                | MEDIUM8           | 6                                                               | 1                      | 2                               | 2                                     | 1245.465               |                       |                        |
| Special Requirements | Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br>MSA Scheduled Aperture PA 158.5746 to 158.5746 Degrees (V3 20.0 to 20.0)<br><br>Group Observations 8, 9, Non-interruptible |                           |                      |                   |                                                                 |                        |                                 |                                       |                        |                       |                        |

# Proposal 5105 - Observation 10 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                   |                                                                                                                                                                                                                                        |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|-------------------------------------------------------------------------------------|-----------------|------------------|--------------------------|-----------------|---------------------|--------------------|----------------------------------------|------------|-----------------------------|
| Observation       | Proposal 5105, Observation 10: Deep-epoch5 NIRCam                                                                                                                                                                                      |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    | Tue Feb 24 20:01:50 GMT 2026           |            |                             |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                             |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Observing Template: NIRCam Imaging                                                                                                                                                                                                     |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                         |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Diagnostics       | (Deep-epoch5 NIRCam (Obs 10)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 10:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 10:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 10:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 10:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 10:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Fixed Targets     | #                                                                                                                                                                                                                                      | Name                |             | Target Coordinates                                                                  |                 |                  | Targ. Coord. Corrections |                 |                     | Miscellaneous      |                                        |            |                             |
|                   | (1)                                                                                                                                                                                                                                    | NEXUS-Center        |             | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                                |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   |                                                                                                                                                                                                                                        |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Template          | NIRCam Imaging                                                                                                                                                                                                                         |                     |             |                                                                                     |                 | MIRI Imaging     |                          |                 |                     |                    |                                        |            |                             |
|                   | Module: ALL                                                                                                                                                                                                                            |                     |             |                                                                                     |                 | Subarray: FULL   |                          |                 |                     |                    |                                        |            |                             |
|                   | Subarray: FULL                                                                                                                                                                                                                         |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Target Placement: Module gap (large extended source)                                                                                                                                                                                   |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Mosaic            | Rows                                                                                                                                                                                                                                   | Columns             |             | Row Overlap %                                                                       |                 | Column Overlap % |                          | Row shift (deg) |                     | Column shift (deg) |                                        | Tile Order |                             |
|                   | 3                                                                                                                                                                                                                                      | 2                   |             | 3.0                                                                                 |                 | 3.0              |                          | 0.0             |                     | 0.0                |                                        | DEFAULT    |                             |
| Dithers           | #                                                                                                                                                                                                                                      | Primary Dither Type |             |                                                                                     | Primary Dithers |                  | Dither Size              |                 | Subpixel Positions  |                    | Coordinated Parallel Subpixel Selector |            | Dither Direct Images Primes |
|                   | 1                                                                                                                                                                                                                                      | FULLBOX             |             |                                                                                     | 2TIGHTGAPS      |                  |                          |                 | 1                   |                    | NIRCam Only                            |            | NO_DITHERING                |
| Spectral Elements | NIRCam Imaging                                                                                                                                                                                                                         | Short Filter        | Long Filter | Readout Pattern                                                                     | Groups/Int      | Integrations/Exp | Total Integrations       | Total Dithers   | Total Exposure Time | Optional ETC ID    |                                        |            |                             |
|                   | 1                                                                                                                                                                                                                                      | F200W               | F444W       | SHALLOW4                                                                            | 6               | 1                | 2                        | 2               | 622.733             |                    |                                        |            |                             |

# Proposal 5105 - Observation 10 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | MIRI Imaging                                                                                                                                                                                                  | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|------------|------------------|----------------|--------|---------------|--------------------|---------------------|-----------------|
|                      | 1                                                                                                                                                                                                             | F1000W | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
| Special Requirements | <p>Between Dates 25-JAN-2026:00:00:00 and 01-FEB-2026:00:00:00</p> <p>Group Visits within 3 Days</p> <p>Visits Same PA</p> <p>No Parallel Attachments</p> <p>Group Observations 10, 11, Non-interruptible</p> |        |                 |            |                  |                |        |               |                    |                     |                 |

# Proposal 5105 - Observation 11 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|               |                                                                                              |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|---------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|---------------------|-----------------|--------------------------|------------------|--------------------|---------------------|-----------------|------------------------------|
| Observation   | Proposal 5105, Observation 11: Deep-epoch5 NIRSpec MOS                                       |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 | Tue Feb 24 20:01:50 GMT 2026 |
|               | Diagnostic Status: Warning                                                                   |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Observing Template: NIRSpec MultiObject Spectroscopy                                         |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Coordinated Parallel Template(s): NIRCам Imaging                                             |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Diagnostics   | (Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 11:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 11:3) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 11:4) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Fixed Targets | #                                                                                            | Name                                                                              | Target Coordinates                                                                  |        |                     |                 | Targ. Coord. Corrections |                  |                    | Miscellaneous       |                 |                              |
|               | (9)                                                                                          | msa_catalog_deep_ep5                                                              | RA: 17 53 50.4509 (268.4602121d)<br>Dec: +65 11 54.74 (65.19854d)<br>Equinox: J2000 |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Comments:<br>Description=[]                                                                  |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Acquisition   | NIRSpec MultiObject Spectroscopy                                                             | Reference Star Bin                                                                | Target                                                                              | Filter | MSA Configuration   | Readout Pattern | Groups/Int               | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |                              |
|               | 1                                                                                            | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 3 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6      | 3                        | 1                | 4                  | 687.153             |                 |                              |
|               | 2                                                                                            | Filter: F110W; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                                                                                | F110W  | Auto Acq MSA Config | NRSRAPID        | 3                        | 1                | 4                  | 171.788             |                 |                              |
|               | 3                                                                                            | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6      | 3                        | 1                | 4                  | 687.153             |                 |                              |
|               | 4                                                                                            | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6      | 3                        | 1                | 4                  | 687.153             |                 |                              |

# Proposal 5105 - Observation 11 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                 |                                                  |       |            |           |           |                |       |            |           |           |
|-----------------|--------------------------------------------------|-------|------------|-----------|-----------|----------------|-------|------------|-----------|-----------|
| Template        | NIRSpec MultiObject Spectroscopy                 |       |            |           |           | NIRCam Imaging |       |            |           |           |
|                 | TA Method: MSATA                                 |       |            |           |           | Module: ALL    |       |            |           |           |
|                 | HFF Readout Mode: false                          |       |            |           |           | Subarray: FULL |       |            |           |           |
|                 | Obtain Confirmation Images: No                   |       |            |           |           |                |       |            |           |           |
|                 | Science Aperture: MSA Center                     |       |            |           |           |                |       |            |           |           |
|                 | Primary Candidate List: Primary (227 sources)    |       |            |           |           |                |       |            |           |           |
|                 | Filler Candidate List: Secondary (19301 sources) |       |            |           |           |                |       |            |           |           |
| Reference Stars | Spectral Overlap Map: jwst-nirspec-prism         |       |            |           |           |                |       |            |           |           |
|                 | Spectral Overlap Threshold: 1.5                  |       |            |           |           |                |       |            |           |           |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 1                                                | 12210 | 268.560811 | 65.195075 | 23.24     | 1              | 19697 | 268.507018 | 65.230060 | 24.48     |
|                 | 1                                                | 17022 | 268.510212 | 65.249253 | 23.53     | 1              | 19770 | 268.494548 | 65.229005 | 24.14     |
|                 | 1                                                | 17108 | 268.517072 | 65.248897 | 24.37     | 1              | 22686 | 268.588094 | 65.214919 | 23.21     |
|                 | 1                                                | 19295 | 268.526722 | 65.231960 | 23.47     | 1              | 24827 | 268.476593 | 65.204905 | 23.74     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 2                                                | 3621  | 268.455547 | 65.147241 | 21.58     | 2              | 9962  | 268.577386 | 65.184951 | 21.03     |
|                 | 2                                                | 3731  | 268.453234 | 65.148268 | 21.71     | 2              | 11034 | 268.470432 | 65.189125 | 21.90     |
|                 | 2                                                | 5016  | 268.549348 | 65.159349 | 21.08     | 2              | 11399 | 268.477912 | 65.190847 | 21.79     |
|                 | 2                                                | 9527  | 268.507806 | 65.183118 | 20.34     | 2              | 11455 | 268.547391 | 65.192344 | 19.95     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 3                                                | 4024  | 268.390536 | 65.149941 | 24.10     | 3              | 11688 | 268.339051 | 65.192271 | 24.28     |
|                 | 3                                                | 4184  | 268.326564 | 65.151041 | 24.70     | 3              | 12775 | 268.361983 | 65.196940 | 23.14     |
|                 | 3                                                | 4812  | 268.315738 | 65.156835 | 24.17     | 3              | 25317 | 268.422974 | 65.200023 | 22.58     |
|                 | 3                                                | 5774  | 268.428733 | 65.164391 | 24.38     | 3              | 25522 | 268.414789 | 65.199150 | 24.41     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 4                                                | 16158 | 268.372076 | 65.255272 | 23.40     | 4              | 20847 | 268.451331 | 65.224156 | 23.91     |
|                 | 4                                                | 16325 | 268.452825 | 65.255678 | 23.50     | 4              | 21424 | 268.446357 | 65.220460 | 22.44     |
|                 | 4                                                | 20249 | 268.436871 | 65.227002 | 23.84     | 4              | 22386 | 268.341644 | 65.216420 | 22.70     |
|                 | 4                                                | 20358 | 268.348676 | 65.226530 | 22.97     | 4              | 23811 | 268.330123 | 65.209536 | 22.57     |
| Dithers         | #                                                |       |            |           |           | Dither Type    |       |            |           |           |
|                 | 1                                                |       |            |           |           | NONE           |       |            |           |           |



# Proposal 5105 - Observation 11 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | NIRSpec MultiObject Spectroscopy                                                                                                                                                                        | Exposure Specification | MSA Configuration | Nod Pattern       | Pointing                                               | Aperture PA        | Dispersion Offset (Shutters) | Cross-Dispersion Offset (Shutters) | Total Dithers       | Total Integrations | Total Exposure Time |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|-------------------|--------------------------------------------------------|--------------------|------------------------------|------------------------------------|---------------------|--------------------|---------------------|
|                      | 1                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep5pt1       | 2 Shutter Slitlet | 268.54303329166 67 Degrees 65.222521666666 67 Degrees  | 98.649845226009 06 |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 2                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep5pt2       | 2 Shutter Slitlet | 268.51866054166 663 Degrees 65.165609722222 23 Degrees | 98.627616252592 45 |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 3                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep5pt3       | 2 Shutter Slitlet | 268.38073933333 334 Degrees 65.174906388888 9 Degrees  | 98.502348121552 4  |                              |                                    | 2                   | 2                  | 1283.822            |
|                      | 4                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep5pt4       | 2 Shutter Slitlet | 268.40187354166 67 Degrees 65.233989444444 45 Degrees  | 98.521614397790 29 |                              |                                    | 2                   | 2                  | 1283.822            |
| Spectral Elements    | NIRCam Imaging                                                                                                                                                                                          | Short Filter           | Long Filter       | Readout Pattern   | Groups/Int                                             | Integrations/Exp   | Total Integrations           | Total Dithers                      | Total Exposure Time | Optional ETC ID    |                     |
|                      | 1                                                                                                                                                                                                       | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                  | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 2                                                                                                                                                                                                       | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                  | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 3                                                                                                                                                                                                       | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                  | 2                            | 2                                  | 1245.465            |                    |                     |
|                      | 4                                                                                                                                                                                                       | F210M                  | F360M             | MEDIUM8           | 6                                                      | 1                  | 2                            | 2                                  | 1245.465            |                    |                     |
| Special Requirements | Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br>MSA Scheduled Aperture PA 98.5746 to 98.5746 Degrees (V3 320.0 to 320.0)<br><br>Group Observations 10, 11, Non-interruptible |                        |                   |                   |                                                        |                    |                              |                                    |                     |                    |                     |

# Proposal 5105 - Observation 12 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                   |                                                                                                                                                                                                                                        |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|-------------------------------------------------------------------------------------|-----------------|------------------|--------------------------|-----------------|---------------------|--------------------|----------------------------------------|------------|-----------------------------|
| Observation       | Proposal 5105, Observation 12: Deep-epoch6 NIRCam                                                                                                                                                                                      |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    | Tue Feb 24 20:01:50 GMT 2026           |            |                             |
|                   | Diagnostic Status: Warning                                                                                                                                                                                                             |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Observing Template: NIRCam Imaging                                                                                                                                                                                                     |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                         |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Diagnostics       | (Deep-epoch6 NIRCam (Obs 12)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified. |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 12:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 12:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 12:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 12:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | (Visit 12:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Fixed Targets     | #                                                                                                                                                                                                                                      | Name                |             | Target Coordinates                                                                  |                 |                  | Targ. Coord. Corrections |                 |                     | Miscellaneous      |                                        |            |                             |
|                   | (1)                                                                                                                                                                                                                                    | NEXUS-Center        |             | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                                |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   |                                                                                                                                                                                                                                        |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Template          | NIRCam Imaging                                                                                                                                                                                                                         |                     |             |                                                                                     |                 | MIRI Imaging     |                          |                 |                     |                    |                                        |            |                             |
|                   | Module: ALL                                                                                                                                                                                                                            |                     |             |                                                                                     |                 | Subarray: FULL   |                          |                 |                     |                    |                                        |            |                             |
|                   | Subarray: FULL                                                                                                                                                                                                                         |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
|                   | Target Placement: Module gap (large extended source)                                                                                                                                                                                   |                     |             |                                                                                     |                 |                  |                          |                 |                     |                    |                                        |            |                             |
| Mosaic            | Rows                                                                                                                                                                                                                                   | Columns             |             | Row Overlap %                                                                       |                 | Column Overlap % |                          | Row shift (deg) |                     | Column shift (deg) |                                        | Tile Order |                             |
|                   | 3                                                                                                                                                                                                                                      | 2                   |             | 3.0                                                                                 |                 | 3.0              |                          | 0.0             |                     | 0.0                |                                        | DEFAULT    |                             |
| Dithers           | #                                                                                                                                                                                                                                      | Primary Dither Type |             |                                                                                     | Primary Dithers |                  | Dither Size              |                 | Subpixel Positions  |                    | Coordinated Parallel Subpixel Selector |            | Dither Direct Images Primes |
|                   | 1                                                                                                                                                                                                                                      | FULLBOX             |             |                                                                                     | 2TIGHTGAPS      |                  |                          |                 | 1                   |                    | NIRCam Only                            |            | NO_DITHERING                |
| Spectral Elements | NIRCam Imaging                                                                                                                                                                                                                         | Short Filter        | Long Filter | Readout Pattern                                                                     | Groups/Int      | Integrations/Exp | Total Integrations       | Total Dithers   | Total Exposure Time | Optional ETC ID    |                                        |            |                             |
|                   | 1                                                                                                                                                                                                                                      | F200W               | F444W       | SHALLOW4                                                                            | 6               | 1                | 2                        | 2               | 622.733             |                    |                                        |            |                             |

Proposal 5105 - Observation 12 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | MIRI Imaging                                                                                                                                                                               | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|-----------------|------------|------------------|----------------|--------|---------------|--------------------|---------------------|-----------------|
|                      | 1                                                                                                                                                                                          | F1000W | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
| Special Requirements | Between Dates 26-MAR-2026:00:00:00 and 02-APR-2026:00:00:00<br>Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br><br>Group Observations 12, 13, Non-interruptible |        |                 |            |                  |                |        |               |                    |                     |                 |

# Proposal 5105 - Observation 13 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|               |                                                                                              |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|---------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------|---------------------|-----------------|--------------------------|------------------|--------------------|---------------------|-----------------|------------------------------|
| Observation   | Proposal 5105, Observation 13: Deep-epoch6 NIRSpec MOS                                       |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 | Tue Feb 24 20:01:50 GMT 2026 |
|               | Diagnostic Status: Warning                                                                   |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Observing Template: NIRSpec MultiObject Spectroscopy                                         |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Coordinated Parallel Template(s): NIRCам Imaging                                             |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Diagnostics   | (Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 13:2) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 13:3) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | (Visit 13:4) Warning (Form): Overheads are provisional until the Visit Planner has been run. |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Fixed Targets | #                                                                                            | Name                                                                              | Target Coordinates                                                                  |        |                     |                 | Targ. Coord. Corrections |                  |                    | Miscellaneous       |                 |                              |
|               | (10)                                                                                         | msa_catalog_deep_ep6                                                              | RA: 17 53 50.4528 (268.4602200d)<br>Dec: +65 11 54.72 (65.19853d)<br>Equinox: J2000 |        |                     |                 |                          |                  |                    |                     |                 |                              |
|               | Comments:<br>Description=[]                                                                  |                                                                                   |                                                                                     |        |                     |                 |                          |                  |                    |                     |                 |                              |
| Acquisition   | NIRSpec MultiObject Spectroscopy                                                             | Reference Star Bin                                                                | Target                                                                              | Filter | MSA Configuration   | Readout Pattern | Groups/Int               | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID |                              |
|               | 1                                                                                            | Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPID        | 3                        | 1                | 4                  | 171.788             |                 |                              |
|               | 2                                                                                            | Filter: F140X; Readout: NRSRAPID; 8 sources in 4 quads; [ Optimal TA Accuracy ]   | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPID        | 3                        | 1                | 4                  | 171.788             |                 |                              |
|               | 3                                                                                            | Filter: F140X; Readout: NRSRAPID; 8 sources in 3 quads; [ Optimal TA Accuracy ]   | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPID        | 3                        | 1                | 4                  | 171.788             |                 |                              |
|               | 4                                                                                            | Filter: F140X; Readout: NRSRAPIDD6; 8 sources in 4 quads; [ Optimal TA Accuracy ] | SAME                                                                                | F140X  | Auto Acq MSA Config | NRSRAPIDD6      | 3                        | 1                | 4                  | 687.153             |                 |                              |

# Proposal 5105 - Observation 13 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                 |                                                  |       |            |           |           |                |       |            |           |           |
|-----------------|--------------------------------------------------|-------|------------|-----------|-----------|----------------|-------|------------|-----------|-----------|
| Template        | NIRSpec MultiObject Spectroscopy                 |       |            |           |           | NIRCam Imaging |       |            |           |           |
|                 | TA Method: MSATA                                 |       |            |           |           | Module: ALL    |       |            |           |           |
|                 | HFF Readout Mode: false                          |       |            |           |           | Subarray: FULL |       |            |           |           |
|                 | Obtain Confirmation Images: No                   |       |            |           |           |                |       |            |           |           |
|                 | Science Aperture: MSA Center                     |       |            |           |           |                |       |            |           |           |
|                 | Primary Candidate List: Primary (215 sources)    |       |            |           |           |                |       |            |           |           |
|                 | Filler Candidate List: Secondary (18527 sources) |       |            |           |           |                |       |            |           |           |
| Reference Stars | Spectral Overlap Map: jwst-nirspec-prism         |       |            |           |           |                |       |            |           |           |
|                 | Spectral Overlap Threshold: 1.5                  |       |            |           |           |                |       |            |           |           |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 1                                                | 14901 | 268.449142 | 65.261192 | 21.39     | 1              | 19078 | 268.382655 | 65.233048 | 20.92     |
|                 | 1                                                | 15579 | 268.413644 | 65.258562 | 21.42     | 1              | 20496 | 268.462525 | 65.225153 | 22.17     |
|                 | 1                                                | 16647 | 268.523209 | 65.251567 | 20.87     | 1              | 20616 | 268.441897 | 65.222079 | 21.68     |
|                 | 1                                                | 18602 | 268.400664 | 65.235951 | 21.29     | 1              | 23221 | 268.436243 | 65.213319 | 22.10     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 2                                                | 10029 | 268.555351 | 65.185164 | 21.37     | 2              | 20059 | 268.584582 | 65.227795 | 22.27     |
|                 | 2                                                | 10930 | 268.510877 | 65.188622 | 21.16     | 2              | 21577 | 268.613939 | 65.219790 | 22.84     |
|                 | 2                                                | 12136 | 268.499536 | 65.194528 | 20.88     | 2              | 22160 | 268.613982 | 65.217623 | 22.37     |
|                 | 2                                                | 16403 | 268.517220 | 65.201217 | 22.13     | 2              | 24760 | 268.585946 | 65.204703 | 21.16     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 3                                                | 1048  | 268.451426 | 65.129462 | 21.99     | 3              | 6939  | 268.465117 | 65.171335 | 21.04     |
|                 | 3                                                | 1315  | 268.479539 | 65.132566 | 20.61     | 3              | 8212  | 268.485870 | 65.176986 | 22.22     |
|                 | 3                                                | 1920  | 268.513736 | 65.136101 | 22.31     | 3              | 9589  | 268.486441 | 65.182937 | 20.77     |
|                 | 3                                                | 6107  | 268.514485 | 65.167020 | 21.19     | 3              | 11034 | 268.470432 | 65.189125 | 21.87     |
|                 | Visit                                            | ID    | RA         | Dec       | Magnitude | Visit          | ID    | RA         | Dec       | Magnitude |
|                 | 4                                                | 6340  | 268.395235 | 65.169336 | 22.41     | 4              | 12827 | 268.401334 | 65.196716 | 23.35     |
|                 | 4                                                | 7352  | 268.397876 | 65.173001 | 22.60     | 4              | 20358 | 268.348676 | 65.226530 | 22.97     |
|                 | 4                                                | 10233 | 268.325579 | 65.186103 | 23.27     | 4              | 22386 | 268.341644 | 65.216420 | 22.70     |
|                 | 4                                                | 10322 | 268.315194 | 65.185495 | 22.80     | 4              | 25317 | 268.422974 | 65.200023 | 22.58     |
| Dithers         | #                                                |       |            |           |           | Dither Type    |       |            |           |           |
|                 | 1                                                |       |            |           |           | NONE           |       |            |           |           |

# Proposal 5105 - Observation 13 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | NIRSpec MultiObject Spectroscopy                                                                                                                                                                        | Exposure Specification | MSA Configuration | Nod Pattern       | Pointing                                               | Aperture PA        | Dispersion Offset (Shutters) | Cross-Dispersion Offset (Shutters) | Total Dithers       | Total Integrations | Total Exposure Time |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|-------------------|--------------------------------------------------------|--------------------|------------------------------|------------------------------------|---------------------|--------------------|---------------------|
|                      | 1                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep6pt1       | 2 Shutter Slitlet | 268.44564554166 664 Degrees 65.240786666666 66 Degrees | 36.561424671399 79 |                              |                                    | 2                   | 4                  | 2217.511            |
|                      | 2                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep6pt2       | 2 Shutter Slitlet | 268.5616695 Degrees 65.205715833333 33 Degrees         | 36.666661817140 9  |                              |                                    | 2                   | 4                  | 2217.511            |
|                      | 3                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep6pt3       | 2 Shutter Slitlet | 268.47424804166 667 Degrees 65.157130000000 01 Degrees | 36.587214073548 3  |                              |                                    | 2                   | 4                  | 2217.511            |
|                      | 4                                                                                                                                                                                                       | 1 (PRISM/CLEAR)        | c1 : ep6pt4       | 2 Shutter Slitlet | 268.36039116666 666 Degrees 65.193169444444 45 Degrees | 36.483958904801 09 |                              |                                    | 2                   | 4                  | 2217.511            |
| Spectral Elements    | NIRCam Imaging                                                                                                                                                                                          | Short Filter           | Long Filter       | Readout Pattern   | Groups/Int                                             | Integrations/Exp   | Total Integrations           | Total Dithers                      | Total Exposure Time | Optional ETC ID    |                     |
|                      | 1                                                                                                                                                                                                       | F210M                  | F360M             | SHALLOW4          | 10                                                     | 2                  | 4                            | 2                                  | 2125.88             |                    |                     |
|                      | 2                                                                                                                                                                                                       | F210M                  | F360M             | SHALLOW4          | 10                                                     | 2                  | 4                            | 2                                  | 2125.88             |                    |                     |
|                      | 3                                                                                                                                                                                                       | F210M                  | F360M             | SHALLOW4          | 10                                                     | 2                  | 4                            | 2                                  | 2125.88             |                    |                     |
|                      | 4                                                                                                                                                                                                       | F210M                  | F360M             | SHALLOW4          | 10                                                     | 2                  | 4                            | 2                                  | 2125.88             |                    |                     |
| Special Requirements | Group Visits within 3 Days<br>Visits Same PA<br>No Parallel Attachments<br>MSA Scheduled Aperture PA 36.5746 to 36.5746 Degrees (V3 258.0 to 258.0)<br><br>Group Observations 12, 13, Non-interruptible |                        |                   |                   |                                                        |                    |                              |                                    |                     |                    |                     |

# Proposal 5105 - Observation 14 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                                                                         |                                                                                                                                                                                                                         |              |                                                                                     |                          |               |                              |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------------------------------------------------------|--------------------------|---------------|------------------------------|
| Observation                                                             | Proposal 5105, Observation 14: Wide-epoch1 WFSS-2                                                                                                                                                                       |              |                                                                                     |                          |               | Tue Feb 24 20:01:50 GMT 2026 |
|                                                                         | Diagnostic Status: Warning                                                                                                                                                                                              |              |                                                                                     |                          |               |                              |
| Observing Template: NIRCam Wide Field Slitless Spectroscopy             |                                                                                                                                                                                                                         |              |                                                                                     |                          |               |                              |
| Coordinated Parallel Template(s): MIRI Imaging                          |                                                                                                                                                                                                                         |              |                                                                                     |                          |               |                              |
| Diagnostics                                                             | (Wide-epoch1 WFSS-2 (Obs 14)) Warning (Form): Use of only one of GRISMR or GRISMC may result in spectral overlap from multiple sources that can't be corrected. Users should address this issue in their proposal text. |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:6) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:7) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:8) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:9) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                            |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:10) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:11) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:12) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:13) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:14) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:15) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:16) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:17) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:18) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:19) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:20) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:21) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:22) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:23) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:24) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:25) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
|                                                                         | (Visit 14:26) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                           |              |                                                                                     |                          |               |                              |
| Fixed Targets                                                           | (Wide-epoch1 WFSS-2 (Obs 14)) Informational (Form): Target Placement is a required field.                                                                                                                               |              |                                                                                     |                          |               |                              |
|                                                                         | #                                                                                                                                                                                                                       | Name         | Target Coordinates                                                                  | Targ. Coord. Corrections | Miscellaneous |                              |
|                                                                         | (1)                                                                                                                                                                                                                     | NEXUS-Center | RA: 17 53 51.0000 (268.4625000d)<br>Dec: +65 11 57.00 (65.19917d)<br>Equinox: J2000 |                          |               |                              |
| Comments:<br>Category=Unidentified<br>Description=[High Latitude Field] |                                                                                                                                                                                                                         |              |                                                                                     |                          |               |                              |

# Proposal 5105 - Observation 14 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                        |                                                                                                                                                                                                           |                     |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|-----------------|------------------|------------------|--------------------|---------------------|--------------------|-------------------------|-------------------------|---------------|
| Template               | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                   |                     |                 |                 |                  |                  |                    | MIRI Imaging        |                    |                         |                         |               |
|                        | Module: ALL                                                                                                                                                                                               |                     |                 |                 |                  |                  |                    | Subarray: FULL      |                    |                         |                         |               |
|                        | Subarray: FULL                                                                                                                                                                                            |                     |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |
|                        | Grism (Long Wavelength): GRISMR                                                                                                                                                                           |                     |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |
|                        | Show partial spectra region in Aladin: false                                                                                                                                                              |                     |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |
| Target Placement: null |                                                                                                                                                                                                           |                     |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |
| Mosaic                 | Rows                                                                                                                                                                                                      | Columns             | Row Overlap %   |                 | Column Overlap % |                  | Row shift (deg)    |                     | Column shift (deg) |                         | Tile Order              |               |
|                        | 9                                                                                                                                                                                                         | 4                   | 5.0             |                 | 5.0              |                  | 0.0                |                     | 0.0                |                         | HILBERT_CURVE           |               |
| Dithers                | #                                                                                                                                                                                                         | Primary Dither Type |                 |                 |                  | Primary Dithers  |                    |                     |                    | Subpixel Positions      |                         |               |
|                        | 1                                                                                                                                                                                                         | INTRAMODULEX        |                 |                 |                  | 2                |                    |                     |                    | NONE                    |                         |               |
| Direct Image           | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                   | Short Filter        | Long Filter     | Readout Pattern | Groups/Int       | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID    | Grism (Long Wavelength) | Exposure Type           | Total Dithers |
|                        | 1                                                                                                                                                                                                         | F150W               | F356W           | SHALLOW4        | 6                | 1                | 1                  | 311.366             |                    | GRISMR                  | Direct Image            | 1             |
|                        | 2                                                                                                                                                                                                         | F090W               | F444W           | SHALLOW4        | 6                | 1                | 1                  | 311.366             |                    | GRISMR                  | Direct Image            | 1             |
| Spectral Elements      | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                   | Short Filter        | Long Filter     | Readout Pattern | Groups/Int       | Integrations/Exp | Total Integrations | Total Exposure Time | Optional ETC ID    | Grism (Long Wavelength) | Exposure Type           | Total Dithers |
|                        | 1                                                                                                                                                                                                         | F115W               | F322W2          | SHALLOW4        | 6                | 1                | 2                  | 622.733             |                    | GRISMR                  | Grism (Long Wavelength) | 2             |
|                        | 2                                                                                                                                                                                                         | F200W               | F444W           | MEDIUM8         | 6                | 1                | 2                  | 1245.465            |                    | GRISMR                  | Grism (Long Wavelength) | 2             |
|                        | 3                                                                                                                                                                                                         | F090W               | F444W           | SHALLOW4        | 6                | 1                | 2                  | 622.733             |                    |                         | Out of Field            | 2             |
| Spectral Elements      | MIRI Imaging                                                                                                                                                                                              | Filter              | Readout Pattern | Groups/Int      | Integrations/Exp | Exposures/Dith   | Dither             | Total Dithers       | Total Integrations | Total Exposure Time     | Optional ETC ID         |               |
|                        | 1                                                                                                                                                                                                         | F770W               | FASTR1          | 57              | 2                | 1                |                    | 2                   | 4                  | 638.259                 |                         |               |
|                        | 2                                                                                                                                                                                                         | F770W               | FASTR1          | 57              | 2                | 1                |                    | 1                   | 2                  | 319.13                  |                         |               |
|                        | 3                                                                                                                                                                                                         | F1000W              | FASTR1          | 56              | 4                | 1                |                    | 2                   | 8                  | 1259.868                |                         |               |
|                        | 4                                                                                                                                                                                                         | F1280W              | FASTR1          | 57              | 2                | 1                |                    | 1                   | 2                  | 319.13                  |                         |               |
|                        | 5                                                                                                                                                                                                         | F1280W              | FASTR1          | 57              | 2                | 1                |                    | 2                   | 4                  | 638.259                 |                         |               |
| Special Requirements   | Between Dates 20-MAY-2025:00:00:00 and 30-JUN-2025:00:00:00<br>Sequence Visits within 5 Days<br>Aperture PA Range 173.0 to 173.0 Degrees (V3 173.0 to 173.0)<br>Visits Same PA<br>No Parallel Attachments |                     |                 |                 |                  |                  |                    |                     |                    |                         |                         |               |



# Proposal 5105 - Observation 15 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

|                               |                                                                                                                                                                                                                                        |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------|-----------------|------------|------------------|--------------------|--------------------------|-----------------|-------------------------|-------------------------|------------------------------|--|
| Observation                   | Proposal 5105, Observation 15: Wide-epoch1 WFSS-2 Copy of Tile-6                                                                                                                                                                       |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         | Tue Feb 24 20:01:50 GMT 2026 |  |
|                               | Diagnostic Status: Warning                                                                                                                                                                                                             |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | Observing Template: NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                            |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | Coordinated Parallel Template(s): MIRI Imaging                                                                                                                                                                                         |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
| Comments: WOPR copy of tile 6 |                                                                                                                                                                                                                                        |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
| Diagnostics                   | (Wide-epoch1 WFSS-2 Copy of Tile-6 (Obs 15)) Warning (Form): Use of only one of GRISMR or GRISMC may result in spectral overlap from multiple sources that can't be corrected. Users should address this issue in their proposal text. |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | (Visit 15:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.                                                                                                                                           |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | (Wide-epoch1 WFSS-2 Copy of Tile-6 (Obs 15)) Informational (Form): Target Placement is a required field.                                                                                                                               |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
| Fixed Targets                 | #                                                                                                                                                                                                                                      | Name                | Target Coordinates                                                                 |                 |            |                  |                    | Targ. Coord. Corrections |                 |                         | Miscellaneous           |                              |  |
|                               | (5)                                                                                                                                                                                                                                    | NEXUS-Center-Tile-6 | RA: 17 55 6.7769 (268.7782371d)<br>Dec: +65 10 43.06 (65.17863d)<br>Equinox: J2000 |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | Comments:<br>Category=Unidentified<br>Description=[High Latitude Field]                                                                                                                                                                |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               |                                                                                                                                                                                                                                        |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
| Template                      | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                                                |                     |                                                                                    |                 |            |                  |                    | MIRI Imaging             |                 |                         |                         |                              |  |
|                               | Module: ALL                                                                                                                                                                                                                            |                     |                                                                                    |                 |            |                  |                    | Subarray: FULL           |                 |                         |                         |                              |  |
|                               | Subarray: FULL                                                                                                                                                                                                                         |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | Grism (Long Wavelength): GRISMR                                                                                                                                                                                                        |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | Show partial spectra region in Aladin: false                                                                                                                                                                                           |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
| Dithers                       | Target Placement: null                                                                                                                                                                                                                 |                     |                                                                                    |                 |            |                  |                    |                          |                 |                         |                         |                              |  |
|                               | #                                                                                                                                                                                                                                      | Primary Dither Type |                                                                                    |                 |            |                  | Primary Dithers    |                          |                 | Subpixel Positions      |                         |                              |  |
|                               | 1                                                                                                                                                                                                                                      | INTRAMODULEX        |                                                                                    |                 |            |                  | 2                  |                          |                 | NONE                    |                         |                              |  |
| Direct Image                  | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                                                | Short Filter        | Long Filter                                                                        | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time      | Optional ETC ID | Grism (Long Wavelength) | Exposure Type           | Total Dithers                |  |
|                               | 1                                                                                                                                                                                                                                      | F150W               | F356W                                                                              | SHALLOW4        | 6          | 1                | 1                  | 311.366                  |                 | GRISMR                  | Direct Image            | 1                            |  |
|                               | 2                                                                                                                                                                                                                                      | F090W               | F444W                                                                              | SHALLOW4        | 6          | 1                | 1                  | 311.366                  |                 | GRISMR                  | Direct Image            | 1                            |  |
| Spectral Elements             | NIRCam Wide Field Slitless Spectroscopy                                                                                                                                                                                                | Short Filter        | Long Filter                                                                        | Readout Pattern | Groups/Int | Integrations/Exp | Total Integrations | Total Exposure Time      | Optional ETC ID | Grism (Long Wavelength) | Exposure Type           | Total Dithers                |  |
|                               | 1                                                                                                                                                                                                                                      | F115W               | F322W2                                                                             | SHALLOW4        | 6          | 1                | 2                  | 622.733                  |                 | GRISMR                  | Grism (Long Wavelength) | 2                            |  |
|                               | 2                                                                                                                                                                                                                                      | F200W               | F444W                                                                              | MEDIUM8         | 6          | 1                | 2                  | 1245.465                 |                 | GRISMR                  | Grism (Long Wavelength) | 2                            |  |
|                               | 3                                                                                                                                                                                                                                      | F090W               | F444W                                                                              | SHALLOW4        | 6          | 1                | 2                  | 622.733                  |                 |                         | Out of Field            | 2                            |  |

# Proposal 5105 - Observation 15 - NEXUS: the North ecliptic pole EXtragalactic Unified Survey

| Spectral Elements    | MIRI Imaging                                                                            | Filter | Readout Pattern | Groups/Int | Integrations/Exp | Exposures/Dith | Dither | Total Dithers | Total Integrations | Total Exposure Time | Optional ETC ID |
|----------------------|-----------------------------------------------------------------------------------------|--------|-----------------|------------|------------------|----------------|--------|---------------|--------------------|---------------------|-----------------|
|                      | 1                                                                                       | F770W  | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
|                      | 2                                                                                       | F770W  | FASTR1          | 57         | 2                | 1              |        | 1             | 2                  | 319.13              |                 |
|                      | 3                                                                                       | F1000W | FASTR1          | 56         | 4                | 1              |        | 2             | 8                  | 1259.868            |                 |
|                      | 4                                                                                       | F1280W | FASTR1          | 57         | 2                | 1              |        | 1             | 2                  | 319.13              |                 |
|                      | 5                                                                                       | F1280W | FASTR1          | 57         | 2                | 1              |        | 2             | 4                  | 638.259             |                 |
| Special Requirements | Aperture PA Range 154.0 to 158.0 Degrees (V3 154.0 to 158.0)<br>No Parallel Attachments |        |                 |            |                  |                |        |               |                    |                     |                 |